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Review

European medicinal polypores – A modern view on traditional uses

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ABSTRACT

Q14 *Ethnopharmacological relevance:* In particular five polypore species, *i.e. Laetiporus sulphureus*, *Fomes fomentarius*, *Fomitopsis pinicola*, *Piptoporus betulinus*, and *Laricifomes officinalis*, have been widely used in central European folk medicines for the treatment of various diseases, *e.g.* Dysmenorrhoea, haemorrhoids, bladder disorders, pyretic diseases, treatment of coughs, cancer, and rheumatism. Prehistoric artefacts going back to over 5000 years underline the long tradition of using polypores for various applications ranging from food or tinder material to medicinal–spiritual uses as witnessed by two polypore species found among items of Ötzi, the Iceman. The present paper reviews the traditional uses, phytochemistry, and biological activity of the five mentioned polypores.
Materials and methods: All available information on the selected polypore taxa used in traditional folk medicine was collected through evaluation of literature in libraries and searches in online databases using SciFinder and Web of Knowledge.
Results: Mycochemical studies report the presence of many primary (*e.g.* polysaccharides) and secondary metabolites (*e.g.* triterpenes). Crude extracts and isolated compounds show a wide spectrum of biological properties, such as anti-inflammatory, cytotoxic, and antimicrobial activities.
Conclusions: The investigated polypores possess a longstanding ethnomycological tradition in Europe. Here, we compile biological results which highlight their therapeutic value. Moreover, this work provides a solid base for further investigations on a molecular level, both compound- and target-wise.
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1. Introduction

Mushrooms have a long history in disease treatment in various folk medicines such as in Asia, Russia, the USA, Canada, Mexico, and Venezuela (Chang, 1999; Garibay-Orijel et al., 2007; Hobbs, 1995) and are extensively applied in Traditional Chinese Medicine (TCM) up to the present day (Chang, 1999). Especially polypore fungi are incorporated into the pharmacopeia and medicine of indigenous people worldwide. Due to their tough and perennial fruit bodies, these bracket fungi have often been regarded as a source of eternal strength and wisdom. Moreover, polypores have been used in various ways as food, tinder, and commodities. Since indigenous people do not clearly distinguish between medicinal and spiritual applications, polypores have been of high symbolic value since ancient times with positive and negative meanings and thus had a strong impact on human culture including art, literature, and folklore (Blanchette et al., 1992; Comandini et al., 2012; Härkönen, 2002; Kreisel, 1998; Molitoris, 2002). Polypore species belonging to the genus *Ganoderma* are some of the oldest traditional medicines. In particular, *Ganoderma lucidum* has been extensively used in TCM as a tonic for promoting health, perpetual youth, vitality, and longevity (Thyagarajan-Sahu et al., 2011). Many studies on *Ganoderma lucidum* extracts or isolates underline its anti-cancer, anti-androgen, immune-stimulating, anti-diabetic, lipid-lowering and anti-inflammatory activities (Grienke et al., 2011; Paterson, 2006; Ying et al., 1987).

Another example for a polypore species widely used in traditional medicine is *Inonotus obliquus*. In Russia, especially in western Siberia, this polypore is called *chaga*. Since the 16th century it has been used as a folk remedy to treat cancer, diseases of the digestive system, and tuberculosis (Shashkina et al., 2006; Zjawiony, 2004). Recent studies claim its anti-AIDS, anti-aging, blood lipid decreasing, blood pressure lowering, and immune-stimulating effects (Zhong et al., 2009).

Most of the studies focusing on constituents and related health effects of medicinal polypores have been conducted in countries with a longstanding tradition of medical mushrooms. The past years have witnessed a renewed interest in the use of mushrooms from traditional medicine which is accompanied by increasing efforts in establishing their medical properties with modern scientific techniques (Chang et al., 2006; Hobbs, 1995; Lindequist et al., 2010). The medical use of traditional mycological products acclaims popular application in Asia and enjoys an excellent reputation, whereas after the introduction of synthetic drugs in Central Europe mycological traditions and knowledge about the medicinal use of mushrooms have been buried in oblivion. This is remarkable since applications of native European polypore species go back to over 5000 years as witnessed by the Iceman, a prehistoric mummy discovered in the Tyrolean Alps in 1991. He was found with tinder material prepared from *Fomes fomentarius* and two objects derived from the birch polypore, *i.e. Piptoporus betulinus*, which he probably used for medicinal–spiritual purposes (Peintner et al., 1998; Pöder and Peintner, 1999). Several other polypore species have also been used in Central European folk medicine, such as *Laetiporus sulphureus* and *Laricifomes officinalis*. Fruit bodies of *Fomitopsis pinicola* are still in use as ornaments on Tyrolean farmhouses and barns, but the medicinal properties of this polypore have been forgotten.

The main aim of this review article is to provide an overview on the available literature concerning the ethnomycological background, health benefits, and bioactive compounds of the five most important polypore species of the Central European folk medicine. Furthermore, we will also highlight major problems hampering comparability of studies and we will give recommendations for how to obtain reliable and reproducible results.

2. Fungal taxonomy and species delimitation

For the processing of mushrooms, species delimitation is a critical point since it is important to characterize and document the starting material for every study conducted.

For the genus *Laetiporus* this is rather problematic since species have usually not been distinguished properly. For instance *Laetiporus* growing on conifers might be *Laetiporus conifericola* or *Laetiporus huronensis*, whereas *Laetiporus* growing on oaks and *Eucalyptus* sp. could be *Laetiporus gilbertsonii* or *Laetiporus cincinnatus* (Burdsall and Banik, 2001; Lindner and Banik, 2008). Hence, one should rely on DNA-based data only to determine taxa within this genus (Banik et al., 2012; Ota et al., 2009; Vasaitis et al., 2009). The species epithet *Laetiporus sulphureus* has often misleadingly been used for closely related species. Therefore, reports on biological effects and isolated constituents of *Laetiporus sulphurous* must be regarded with caution, as it often remains unclear which taxon of this species complex has been investigated. These problems also apply to *Fomes fomentarius* which comprises at least two species in the USA (McCormick et al., 2013), in addition to at least two cryptic sympatric species with different genotypes in Europe (Judova et al., 2012). Also for *Fomitopsis pinicola* the morphological delimitation from closely related recently described new species such as *Fomitopsis palustris*, *Fomitopsis ochracea*, and *Fomitopsis meliae* can sometimes be critical (Kim et al., 2005, 2007). The genus *Fomitopsis* is typified with *Fomitopsis pinicola* but

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Table 1
Overview on names of the five discussed polypore species.

Current scientific name (incl. authority)	Basionymand synonyms	Common name
Laetiporus sulphureus (Bull.) Murrill, Annls mycol. 18(1/3): 51 (1920)	***Boletus sulphureus*** Bull 1798	Chicken of the woods, chicken polypore, sulphur polypore, sulphur shelf
Fomes fomentarius(L.) Fr., Summa veg. Scand., Section Post. (Stockholm): 321 (1849)	***Boletus fomentarius*** L., Sp. pl. 2: 1176 (1753) ***Polyporus fomentarius*** (L.) Fr. (1821)	Tinder fungus, hoof fungus, tinder conk, tinder polypore, Iceman's fungus
Fomitopsis pinicola (Sw.) P. Karst Meddn Soc. Fauna Flora fenn. 6: 9 (1881)	***Boletus pinicola*** Sw., K. Vetensk.-Acad. Nya Handl. 31: 88 (1810) ***Fomitopsis marginata*** (Pers.) (P. Karst.) (1881) ***Fomes marginatus*** (Pers.) (1849) ***Polyporus pinicola*** (Sw.) Fr (1821)	Red banded polypore
Piptoporus betulinus (Bull.) P. Karst., MeddnSoc. Fauna Flora fenn. 6: 9 (1881)	***Boletus betulinus*** Bull., Herb. France (Paris) 7: pl. 312 (1788) ***Polyporus betulinus*** (Bull.) Fr.(1821) ***Boletus suberosus*** L. (1753)	Birch polypore, birch bracket, razor strop
Laricifomes officinalis (Vill.) Kotl. &PouzarČeskáMykol.**11**(3): 158 (1957)	***Boletus officinalis***Vill. Hist. pl. Dauphiné 3(2): 1041 (1788) ***Fomitopsis officinalis***(Vill.) Bondartsev& Singer (1941) ***Polyporus officinalis*** (Vill.) Fr. (1821)	Conks of larch

Fomitopsis is not a monophyletic genus (Kim et al., 2005). Therefore, the combination *Laricifomes officinalis* has to be applied as the oldest legitimate name for *Fomitopsis officinalis*. About 16 rDNA ITS sequences of *Laricifomes officinalis* are currently available in public DNA databases (NCBI), but sequence divergences between some of them are significantly higher than 98%, indicating the existence of cryptic or geographically distinct taxa. For *Piptoporus betulinus*, species identification is comparatively easy, but also here rDNA ITS sequence divergence indicates the presence of distinct clusters within this taxon.

For standardization reasons the current scientific names of the respective polypore species are used throughout this work. However, synonyms are still in use and have been taken into account for the literature research. Table 1 gives an overview on the names of the five selected polypore species (incl. first authors and references). The ethnomycological background of these selected species is discussed in the following chapter.

3. Ethnomycological background of selected polypores

3.1. *Laetiporus sulphureus* – chicken of the woods

Most polypores cannot be used as food because of their hard and corky texture; but young fruit bodies of *Laetiporus sulphureus* are an exception. Hence, this polypore is also called "chicken polypore" or "chicken of the woods" due to its taste and texture resembling poultry. In certain parts of Germany and North America it is therefore considered a delicacy and it can also be used as a substitute for chicken in a vegetarian diet. However, gastrointestinal problems have been reported after eating this fungus as well as the occurrence of severe adverse effects including allergic reactions, vomiting, and fever (Jordan, 1995; Watling, 1997). *Laetiporus sulphureus* consumption has also been reported to cause hallucinations. It has therefore been assumed that this species might contain alkaloids similar to those found in psychoactive plants (Appleton et al., 1988). However, it is very likely that such hallucinogenic effects might rather be associated with a closely related polypore species.

Besides their benefits as food, *Laetiporus sulphureus* fruit bodies are thought to be capable of regulating the human body, improving health and defending the body against illnesses (Ying et al., 1987). Moreover, in Europe, the fruit bodies have been used for the treatment of pyretic diseases, coughs, gastric cancer, and rheumatism (Matt, 1947; Rios et al., 2012). Burning of *Laetiporus sulphureus* fruit body is presumed to drive away mosquitoes and midges (Ying et al., 1987).

3.2. *Fomes fomentarius* – tinder fungus

Fomes fomentarius was found with the 5000-year-old Iceman who might have used this polypore to make and preserve fire, as first aid kit, as insect repellent, or for spiritual purposes (Peintner et al., 1998; Pöder and Peintner, 1999).

Without doubt, *Fomes fomentarius* was used for cauterization since the times of Hippocrates in the fifth century BC (Peintner and Pöder, 2000; Peintner et al., 1998). Interestingly, cauterization with *Fomes fomentarius* has also been in use by the Okanagan-Colville Indians of British Columbia to cure rheumatism. After pounding and softening it, a piece of the fungus was put on the skin over the affected area and ignited (Hobbs, 1995).

Fomes fomentarius was widely used as a styptic by surgeons, barbers and dentists, and therefore called "agaric of the surgeons" (Buller, 1914; Göpfert, 1982). Furthermore, in European, WestSiberian, and Indian folk medicine, a kind of absorbing dressing made of tinder and some iodine is externally applied to wounds and burns (Mellin, 1791; Saar, 1991; Vaidya and Rabba, 1993). In the German speaking Alpine area, *Fomes fomentarius* was called "Wundschwamm" or "Chirurgenschwamm", and was sold in pharmacies in the form of styptic bandages. This absorbing wound tissue was used by Austrian farmers up to the 19th century (Rutalek, 2002). Moreover, around Easter, *Fomes fomentarius* fruit bodies were used for ritual smoking ceremonies in Germany and Austria (Rutalek, 2002). Similar applications are known from Khanty people in West Siberia who used to burn the fruit bodies to obtain smoke when a person died to avoid any influence of the deceased on the living (Saar, 1991).

Besides these external applications, *Fomes fomentarius* was used as a remedy against dysmenorrhoea, haemorrhoids and bladder disorders, the active substance being "fomitin" (Killermann, 1938). Furthermore, there are reports about the use of *Fomes fomentarius* for pain relief (Ying et al., 1987) and for the treatment of oesophagus, gastric and uterine carcinoma (Wasson, 1969).

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Besides its medicinal use, the tinder fungus was also applied for different commodities. In Styria (Austria), the fruit bodies were taken as bungs for bins and as material for carvings which were placed on the necks of farm animals to protect them from bad luck (Lohwag, 1965). In Germany, Hungary, and in some parts of former Yugoslavia, *Fomes fomentarius* was used for making caps, chest protectors, and other clothing articles. In Germany and former Bohemia, large fruit bodies were often used for decorative purposes such as flower pots (Cordier, 1870). The fungi were placed in cupboards because of their pleasant smell (Killermann, 1938). Big specimens were also used to prevent needles from rusting. In fly fishing, pads of "amadou" were used as excellent absorbent for drying water-logged flies (Peintner and Pöder, 2000; Roussel et al., 2002).

3.3. *Fomitopsis pinicola* – red banded polypore

In traditional medicine, *Fomitopsis pinicola* was used for the treatment of headache, nausea, and liver problems. Moreover, due to their astringent effects the fruit bodies have been used as haemostatics and anti-inflammatory agents (Leeser, 1987). Additionally, *Fomitopsis pinicola* was often used as tinder and styptic as alternative for *Fomes fomentarius* (Leeser, 1987; Montoya et al., 2004). In Mexico, people decorated the nativity scene at Christmas with this polypore (Montoya et al., 2004). Also in Tyrol (Austria), fruit bodies of *Fomitopsis pinicola* are still found as decorative item on farm houses. This decorative application of the fungus indicates that it was also used for protecting the house, and that it was regarded to have spiritual-medicinal properties by native people.

3.4. *Piptoporus betulinus* – birch polypore

Piptoporus betulinus is one of the few edible polypores, at least when the fruiting bodies are young (Wasson, 1969). They have a strong, pleasant odour and an astringent, bitter taste.

This polypore was used for various medicinal purposes before modern medicine superseded many natural healing methods. In Russia, fruit bodies were administered as tea for their anti-fatiguing, soothing, and immune-enhancing properties (Peintner and Pöder, 2000). In Siberia, the Baltic area, and Finland, birch polypore tea was also used for the treatment of various types of cancer. Only young, sterile fruit bodies (without developed hymenial layers) were thought to be effective and it was claimed that these develop on birch trees only under certain environmental conditions, particularly when the trees grow on low ground (Lucas, 1960). In Poland, birch polypore extracts were given orally to female dogs with tumours of the vagina which were observed to completely disappear after five weeks (Utzig, 1957). Stripes of *Piptoporus betulinus* fruit bodies were also used externally as styptic and charcoal of this polypore was appreciated as an antiseptic (Hobbs, 1995; Thoen, 1982). A powder produced from *Piptoporus betulinus* fruit bodies was used as snuff in Austria. Interestingly, similar applications have been reported for Northern America and Siberia, where snuff prepared from the ash of *Piptoporus betulinus* was used as pain reliever (Rutalek, 2002).

Besides the nutritional and medicinal purposes, *Piptoporus betulinus* was also used in many other ways. The velvety surface of the fruit body was traditionally taken as a strop for finishing razor edges (Pegler, 2000; Thoen, 1982). One curious application of the fruit bodies of *Piptoporus betulinus* was reported for people from the Scottish Highlands, who used them as packing material for the back of their circular shields or targets (Marsh, 1973). Moreover, carved *Piptoporus betulinus* fruit bodies were used to protect farm animals from bad luck in Styria (Austria) (Lohwag, 1965).These combined artistic and medicinal–spiritual applications might also be the case for carved fruiting bodies of *Piptoporus betulinus* carried by the Iceman on his journey over the Alps (Peintner et al., 1998; Pöder and Peintner, 1999).

3.5. *Laricifomes officinalis* – conks of larch

Inhabitants of Agaria in Sarmatia, living in an area covering modern Ukraine and Southern Russia, used a polypore that they named *agarikon* to combat illness. It is not completely clear which polypore they used but it was agreed on that it was either *Fomes fomentarius* and/or *Laricifomes officinalis* (Berendes, 1902). In the 1st century AD, the Greek philosopher Dioscorides recorded the name as αγαρικόν. Its use persisted throughout medieval times and it was prescribed as one of the herbal remedies for tuberculosis (Berendes, 1902; Buller, 1914; Weber, 1958).

Laricifomes officinalis fruit bodies have been extensively collected for medicinal purposes throughout the whole Alpine area, nearly leading to the extinction of this rare polypore (Senn-Irlet, 2012).

People collected it to be sold to pharmacies. Therefore, this fungus was called Agaric of Pharmacy or "Apothekerschwamm" in German speaking regions. The fruit bodies were collected, dried and pulverized to a smooth powder. A formula on a hand-written note from 1787 found in a book of an antique pharmacy (Stadtapotheke Peer, Brixen, Italy) in South Tyrol, states that *Laricifomes officinalis* was used as the chief ingredient for "Großer Schwedenbitter", *i.e.* an alcoholic herbal extract used to cure stomach and digestion problems. However, applied in the form of a bitter liquor the intake of *Laricifomes officinalis* was frequently accompanied by diarrhoea, colic and other side effects (Killermann, 1938). In folk medicine, the bitter fruit bodies were traditionally used to treat coughs, gastric cancer, and rheumatism (Airapetova et al., 2010). However, especially in the Ukraine, its main area of application was as an antiperspirant to relieve night sweats associated with fever and tuberculosis. Also in Canada and the United States *Laricifomes officinalis* has been traditionally used for the treatment of tuberculosis, pneumonia, cough, and asthma (Hwang et al., 2013), but also as a poultice for swollen and inflamed areas (Blanchette et al., 1992). To indigenous people of the northwestern coast region of North America and Canada, this fungus is known as "bread of ghosts". It had an important spiritual as well as a medicinal role in their society. Its supernatural powers were intensified through shamanic art, *e.g.* carved fruit bodies as Shaman grave guardians. Such rituals demonstrate the supernatural prestige that polypores had among indigenous people (Blanchette et al., 1992; Smith, 1929).

4. Primary metabolites of selected polypores and their bioactivities

A lot of efforts have been put into research focusing on primary metabolites derived from polypores (Xu et al., 2011a; Zhang et al., 2011). However, in many studies bioactivities are associated with complex multi-component mixtures or particular compound groups, in a speculative manner, without chemical characterization. The most important medically active primary metabolites from fungi comprise high-molecular weight compounds such as polysaccharides, proteins, and polysaccharide–protein complexes. In addition, some pigments and nucleic acids have also been described to be biologically active.

4.1. Polysaccharides

Fungal polysaccharides can be classified as α-glucans (*e.g.* starch, cellulose, or chitin) and β-glucans including their derivatives (Jiang et al., 2010; Moradali et al., 2007). α-Glucans have shown little or no bioactivity, whereas β-glucans are responsible for various biological properties. They are one of the major constituents of the fungal cell wall and consist of a backbone of glucose residues linked by β-(1,3)-glycosidic bonds, often with attached side-chain glucose residues joined by β-(1,6) linkages (Moradali et al., 2007). The frequency of branching varies and thus

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gives an abundance of different types of these metabolites (Mattila et al., 2000; Vannucci et al., 2013). Biological effects of fungal polysaccharides include immune-regulatory (Jiang et al., 2010), anti-tumour (Chen et al., 2008), antiviral (Teplyakova et al., 2012), anti-inflammatory (Moro et al., 2012), antioxidant (Klaus et al., 2013; Sun et al., 2012), and hypoglycaemia activity (Cha et al., 2009; Hwang and Yun, 2010). Remarkably, polysaccharides have no reported adverse effects; on the contrary they help the body to adapt to biological and environmental stress (Jiang et al., 2010).

Laetiporus sulphureus fruit bodies are a rich source of α-(1,3)-D-glucans. Their cell wall contains up to 88% of dry weight of this glucan, whereas in other fungi it is present in an amount of 9–46% only (Wiater et al., 2012). Antioxidant effects have been reported for both, water-soluble and alkali-soluble polysaccharides extracted from *Laetiporus sulphureus* fruit bodies (Klaus et al., 2013; Olennikov et al., 2009a, b). Crude extracellular polysaccharides (EPS) produced from a submerged mycelial culture of *Laetiporus sulphureus*, water extracts of *Fomes fomentarius*, and water as well as alkali extracts of *Fomitopsis pinicola* provoked a hypoglycaemia effect in streptozotocin (STZ)-induced diabetic rats, indicating that these substances could be useful in diabetes mellitus treatment (Hwang and Yun, 2010; Lee, 2005; Lee et al., 2008). Moreover, EPS also inhibit the expression of pro-inflammatory mediators (Jayasooriya et al., 2011) and activate immune-modulating mediators (Seo et al., 2011). In addition, EPS from *Fomes fomentarius* showed an *in vitro* anti-proliferative effect on SGC-7901 human gastric cancer cells in a dose- and time-dependent manner without being cytotoxic. EPS promote the secretion of TNF-α, IFN-γ, and IL-2 by mouse immunocytes and enhance mouse humoral immune response and the phagocytotic activity of macrophages (Gao et al., 2009). Also intracellular polysaccharides (IPS) from *Fomes fomentarius* have shown a direct anti-proliferative effect on human gastric cancer cell lines SGC-7901 and MKN-45 in a dose-dependent manner (Chen et al., 2011).Carboxymethylated α-(1,3)-D-glucans of *Piptoporus betulinus* fruit bodies have shown to exert cytotoxic effects (Wiater et al., 2011). Moreover, mycelium culture extracts of *Laricifomes officinalis* have shown antibacterial properties against Gram-negative bacteria (Sidorenko and Buzoleva, 2012), and antiviral activity against type A influenza virus of birds A/chicken/Kurgan/05/2005 (H5N1) and humans A/Aichi/2/68 (H3N2) (Teplyakova et al., 2012). *Laricifomes officinalis* extracts were also evaluated for their anti-aging potential as an ingredient for cosmetics since they induced a neuromuscular blockade simulating the effect of botox (Santana et al., 2011).

4.2. Proteins

Besides the most extensively studied polysaccharides, bioactive proteins constitute another abundant primary metabolite component in mushrooms (Xu et al., 2011b). Bioactivities related to these proteins include antitumor, antiviral, antimicrobial, antioxidative, and immunomodulatory properties (Kang et al., 1982; Xu et al., 2011b). From a structural perspective, fungal proteins can be categorized as classical proteins/peptides (including enzymes), or lectins, *i.e.* carbohydrate-binding proteins.

These macromolecules are usually isolated by using a combination of affinity and ion-exchange chromatography. Only in rare cases, lectins from the five selected polypore species have been characterized in detail (Konska et al., 1994). However, in studies dealing with lectins from related polypore fungi such as *Polyporus squamosus*, more detailed information on purification and characterization is given (Mo et al., 2000). A *Polyporus squamosus* lectin (PSA) was analysed by using gel filtration chromatography, SDS-polyacrylamide gel electrophoresis, and N-terminal amino acid sequencing. Further experiments revealed insights into blood cell specific agglutinating activities and carbohydrate binding properties with a specificity for terminal α2,6-linked Neu5Ac. Hence, due to these structural specificities, lectins can be considered as especially interesting for cancer research and glycobiological studies (Mo et al., 2000). However, toxicological aspects of these bioactive proteins should not be neglected. Such issues have been addressed for example for a lectin from the shiitake mushroom which showed no acute toxicity in mice up to a concentration of 10,000 mg/kg body weight (Eghianruwa et al., 2011).

4.3. Polysaccharide–protein complexes

Polysaccharides can reach a high level of complexity when they are covalently bound to other conjugate molecules such as polypeptides and proteins. Such polysaccharide–protein complexes or polysaccharopeptides have promising bioactive properties due to their significant immune-stimulatory activity (Cui and Chisti, 2003; Sakagami, 1991). Several clinical trials have already shown the benefits of polysaccharide–protein complexes obtained from edible mushrooms for immune stimulation and cancer treatment without any toxic effect (Gonzaga et al., 2009; Ishii et al., 2011). The best known commercially available examples for this class of primary metabolites are polysaccharide-K (krestin, PSK) and its analogue polysaccharide peptide (PSP) (Cui and Chisti, 2003) obtained from the polypore *Trametes versicolor* (*i.e. Coriolus versicolor*). PSK and PSP are chemically similar and possess similar physiological activity profiles. A protein–polysaccharide fraction (PPF) from fruit bodies of *Laetiporus sulphureus* consisting of 84% polysaccharide and 5% protein exerted antitumor activity against sarcoma 180 in mice (Kang et al., 1982). However, it is important to consider that polysaccharopeptides isolated from different sources of a fungus (fruit body, mycelium, or biomass-free growth medium) differ in structure, composition, and physiological activity (Cui and Chisti, 2003).

4.4. Pigments and other primary metabolites

Recently, a water-soluble melanin–glucan complex (MGC; 80% melanin and 20% β-glucan) was investigated on different microbial pathogens and showed a fungistatic effect against *Candida albicans in vitro*, an antimicrobial effect on *Helicobacter pylori* identical to erythromycin in all concentrations tested, and a high anti-HIV-1 activity in comparison with zidovudine (Retrovir). Furthermore, an insoluble chitin–glucan–melanin complex (ChGMC; 70% chitin, 20% β-glucan, and 10% melanin) has also shown anti-infective properties. Both, MGC and ChGMC showed no toxic properties on blood cells (Seniuk et al., 2011).

Laetiporic acids, *i.e.* non-carotenoid polyenepigments, identified in *Laetiporus sulphureus* fruit bodies have well-known antioxidant properties and their high stability might render them attractive as food dye (Davoli et al., 2005; Weber et al., 2004).

Several bioactive primary metabolites have also been isolated and identified from *Piptoporus betulinus*. For instance, nucleic acids isolated from its fruit bodies have shown to reduce the number of vaccinia virus plaques in chick embryo fibroblast (CEF) tissue culture, an effect attributed to the induction of interferon production *in vivo* (Kandefer-Szerszen et al., 1979).

5. Secondary metabolites of selected polypores

5.1. Data evaluation of available literature dealing with secondary metabolites

An extensive literature search for secondary metabolites of the five polypores under investigation was performed using SciFinder Scholar (Chemical Abstracts Service – http://www.cas.org/products/sfacad/index.html) and ISI Web of Knowledge (Thomson

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Table 2
Polypore derived secondary metabolites including available information on biological properties.

No.	Compound name	Polypore species	Biological activity	Reference(s)
(A) Triterpenoids				
Acids				
1	Eburicoic acid	LS, LO	Anti-cancer, anti-thrombin (n.a.)	Chen et al., (2005), Feng et al. (2010), León et al. (2004), Sheth et al. (1967), Shiono et al. (2005) and X. Wu et al. (2005)
2	Sulfurenic acid	LS, LO	anti-leukaemia, anti-thrombin (n.a.)	Anderson and Epstein (1971), Chen et al. (2005), León et al. (2004), and X. Wu et al. (2005)
3	Fomefficinic acid C	LO	n.g.	Wu et al. (2004)
4	Versisponic acid D	LO	Anti-thrombin	X. Wu et al. (2005)
5	3-O-Acetyleburicoic acid	LS	Anti-leukaemia	León et al. (2004)
6	Pachymic acid	FP	Antimicrobial (n.a.)	Chen et al. (2005), Keller et al. (1996)
7	Fomefficinic acid A	LO	n.g.	Wu et al. (2004)
8	16α-Hydroxyeburiconic acid	FP	Antibacterial	Liu et al. (2010)
9	Fomefficinic acid D	LS, LO	Anti-leukaemia	León et al., (2004), Wu et al. (2004)
10	Versisponic acid C	LS	Anti-leukaemia	León et al. (2004)
11	(+)-Trametenolic acid B	FP	n.g.	Keller et al., (1996), Rösecke and König (1999)
12	15α-Hydroxytrametenolic acid	LS	Anti-leukaemia	León et al. (2004)
13	(3β)-3-(acetyloxy)-Lanosta-8,24-dien-21-oic acid	LS	Anti-leukaemia	León et al. (2004)
14	Tsugaric acid A	FP	Antibacterial	Keller et al., (1996), Petrova et al. (2007), Rösecke and König (1999)
15	Pinicolic acid A	FP	Antibacterial	Keller et al. (1996), Liu et al. (2010), Petrova et al. (2007), Rösecke and König (1999)
16	Pinicolic acid B	FP	n.g.	Rösecke and König (1999)
17	Pinicolic acid E	FP	n.g.	Rösecke and König (2000)
18	Fomitopsic acid	FP	Antimicrobial	Keller et al. (1996)
19	24-Methyl-3-oxo-Lanosta-8,25-dien-21-oic acid	FP	Antibacterial	Liu et al. (2010)
20	Fomitopinic acid A	FP	Anti-inflammatory	Yoshikawa et al. (2005)
21	Fomitopinic acid B	FP	Anti-inflammatory	Yoshikawa et al. (2005)
22	Fomefficinic acid F	LO	n.g.	Wu et al. (2009)
23	Polyporenic acid A	PB	Anti-inflammatory	Kamo et al. (2003)
24	Fomefficinic acid G	LO	n.g.	Wu et al. (2009)
25	(3α,12α,25S)-3-(acetyloxy)-12-hydroxy-24-methylene-Lanost-8-en-26-oic acid	PB	Anti-inflammatory, inhibition of bacterial Hyaluronidase	Wangun et al. (2004)
26	(3α,12α,25S)-3-[(carboxyacetyl)oxy]-12-hydroxy-24-methylene-Lanost-8-en-26-oic acid	PB	Anti-inflammatory	Kamo et al. (2003)
27	(3α,12α,25S)-12-hydroxy-3-(3-methoxy-1,3-dioxopropoxy)-24-methylene-Lanost-8-en-26-oic acid	PB	Anti-inflammatory, inhibition of bacterial hyaluronidase	Wangun et al. (2004)
28	(3α,12α,25S)-3-[(3S)-4-carboxy-3-hydroxy-3-methyl-1-oxobutoxy]-12-hydroxy-24-methylene-Lanost-8-en-26-oic acid	PB	Anti-inflammatory	Kamo et al. (2003)
29	(3α,12α,25S)-12-hydroxy-3-[[(3S)-3-hydroxy-5-methoxy-3-methyl-1,5-dioxopentyl]oxy]-24-methylene-Lanost-8-en-26-oic acid	PB	Anti-inflammatory, inhibition of bacterial hyaluronidase	Kamo et al., (2003), Wangun et al. (2004)
30	(+)-12α,28-Dihydroxy-3α-(3′-hydroxy-3′-methylglutaryloxy)-24-methyllanosta-8,24(31)-dien-26-oic acid	PB	Anti-inflammatory	Kamo et al. (2003)
31	Dehydroeburicoic acid	LO	anti-thrombin (n.a.)	Feng et al. (2010) and X. Wu et al. (2005)
32	Dehydrosulfurenic acid	LO	anti-thrombin (n.a.)	Feng et al. (2010) and X. Wu et al. (2005)
33	Fomefficinic acid B	LO	n.g.	Wu et al. (2004)
34	Dehydroeburiconic acid	LO	Anti-thrombin (n.a.)	Feng et al. (2010) and X. Wu et al. (2005)
35	Polyporenic acid C	PB, FP	Anti-inflammatory, inhibition of bacterial hyaluronidase, antibacterial	Chen et al. (2005), Hybelbauerova et al. (2008), Kamo et al. (2003), Kawagishi et al. (2002), Keller et al. (1996), Liu et al. (2010) and Wangun et al. (2004)
36	(16α)-16-(acetyloxy)-24-methylene-3-oxo-Lanosta-7,9(11)-dien-21-oic acid	FP	Antibacterial	Liu et al. (2010)
37	3-Ketodehydrosulfurenic acid	LO	Anti-thrombin (n.a.)	Anderson et al., (1972), Feng et al. (2010) and X. Wu et al. (2005)
38	Fomefficinic acid E	LO	n.g.	Wu et al. (2004)
39	Pinicolic acid D	FP	n.g.	Rösecke and König (1999)
40	Fomitopsic acid B	FP	n.g.	Rösecke and König (1999)
Esters and lactones				
41	Methyl polyporenate C	PB	n.g.	Bryce et al. (1967)
42	(3α)-Lanosta-8,24-diene-3,21-diol 3-acetate	FP	n.g.	Petrova et al. (2007)
43	3α,12α-dihydroxy-24-methylene-Lanost-8-en-26-oic acid methyl ester	PB	n.g.	Bryce et al. (1967)
44	3α,12α-dihydroxy-24-methylene-Lanost-8-en-26-oic acid methyl ester 3-acetate	PB	n.g.	Bryce et al. (1967)
45	[3β(9Z,12Z),5α,22E]-Ergosta-7,22-dien-3-ol, 9,12-octadecadienoate	FP, FF	n.g.	Rösecke and König (2000)
46	Fungisterollinoleate	FP, FF	n.g.	Rösecke and König (2000)
47	[3β(9Z,12Z)]-Ergosta-7,24(28)-dien-3-ol, 9,12-octadecadienoate	FP, FF	n.g.	Rösecke and König (2000)
48	Fomefficinol A	LO	n.g.	Wu et al. (2009)
49	Fomefficinol B	LO	n.g.	Wu et al. (2009)

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Table 2 (*continued*)

No.	Compound name	Polypore species	Biological activity	Reference(s)
50	Fomlactone A	LO	n.g.	Wu et al. (2009)
51	Fomlactone B	LO	n.g.	Wu et al. (2009)
52	Fomlactone C	LO	n.g.	Wu et al. (2009)
53	Betulin 28-O-acetate	FF	Antitumor	Huang et al. (2012)
Alcohols				
54	24-Ethylcholestan-3β-ol	LS	n.g.	Kac et al. (1984)
55	Δ7-Ergostenol	LS, FF	Antitumor	Coy and Nieto (2009), Huang et al. (2012) and Kac et al. (1984)
56	α-Dihydroergosterol	LO, LS, FP	Antimicrobial (n.a.)	Feng and Yang (2010), Kac et al. (1984), Keller et al. (1996) and Wu et al. (2009)
57	6-Epicerevisterol	FF	Cytotoxic (n.a.)	Zang et al. (2013)
58	(22E,24R)-Ergosta-7,22-diene-3β,5α,6α,9α-tetrol	FF	Cytotoxic (w.a.)	Zang et al. (2013)
59	Cerevisterol	LS, FF	Anti-leukaemia (n.a.), cytotoxic (n.a.)	Coy and Nieto (2009), León et al. (2004) and Zang et al. (2013)
60	Brassicasterol	LS	n.g.	Coy and Nieto (2009)
61	β-Sitosterol	FF	n.g.	Feng and Yang (2010)
62	CAS no. 1444001-92-6	FF	Cytotoxic (n.a.)	Zang et al. (2013)
63	Ergosterol	LS, LO	n.g.	Coy and Nieto (2009), Kac et al., (1984) and Wu et al. (2009)
64	(22E,24R)-Ergosta-7,9(11),22-triene-3β,5α,6β-triol	FF	Cytotoxic (n.a.)	Zang et al. (2013)
65	CAS no. 1444001-91-5	FF	Cytotoxic (w.a.)	Zang et al. (2013)
66	Ergosterol D	FP	Antibacterial	Liu et al. (2010)
67	Dehydroergosterol	LS	n.g.	Coy and Nieto (2009)
68	(3β)-4-Methyl-ergosta-7,14,25-trien-3-ol	LS	n.g.	Coy and Nieto (2009)
69	(3β)-4-methyl-Ergosta-5,7,25-trien-3-ol	LS	n.g.	Coy and Nieto (2009)
70	(3β)-4,4-dimethyl-Ergost-24(28)-en-3-ol	LS	n.g.	Coy and Nieto (2009)
71	Obtusifoldienol	LS, LO	n.g.	Coy and Nieto (2009) and Epstein and van Lear (1966)
72	Eburicodiol	LO	n.g.	Anderson and Epstein (1971)
73	24-Methyleneagnosterol	LS	n.g.	Coy and Nieto (2009)
74	5α-Lanosta-7,9(11),24-triene-3β,21-diol	FP	n.g.	Rösecke and König (1999)
75	Pinicolol B	FP	n.g.	Rösecke and König (1999)
76	(+)-Betulin	FF	Antitumor	Huang et al. (2012)
Ethers and peroxides				
77	CAS no. 1444001-93-7	FF	Cytotoxic (w.a.)	Zang et al. (2013)
78	(5α)-3,3-dimethoxy-Ergosta-7,22-diene	FF	Antitumor	Huang et al. (2012)
79	Ergosterol peroxide	LS, PB, FF	Cytotoxic, anti-leukaemia (n.a.), cytotoxic (n.a.), antitumor	Coy and Nieto (2009), Feng and Yang (2010), Huang et al. (2012), Hybelbauerova et al. (2008), Krzyczkowski et al. (2009), León et al. (2004), Rösecke and König, (2000) and Zang et al. (2013)
80	5,8-epidioxy-Ergosta-6,9(11),22-trien-3β-ol	PB	n.g.	Hybelbauerova et al. (2008)
Aldehydes and ketones				
81	(3β,5α,6β)-3,6-dihydroxy-4,4,14-trimethyl-Pregn-8-en-20-one	LO	n.g.	Anderson et al. (1972)
82	3α-hydroxy-4,4,14α-trimethyl-5α-pregn-8-en-20-one	LO	n.g.	Epstein and van Lear (1966)
83	Eburical	LO	n.g.	Anderson and Epstein (1971)
84	(+)-21-Hydroxylanosta-8,24-dien-3-one	FP	Antimicrobial (n.a.)	Keller et al., (1996) and Rösecke and König (1999)
85	(22E)-Ergosta-7,22-dien-3-one	FF	Antitumor	Du et al. (2011), Huang et al., (2012) and Rösecke and König (2000)
86	5α-Ergost-7-en-3-one	FF	n.g.	Rösecke and König, (2000)
87	Agnosterone	FP	n.g.	Rösecke and König (1999)
88	21-hydroxy-Agnosterone	FP	n.g.	Rösecke and König (1999)
89	Pinicolol C	FP	n.g.	(Rösecke and König, (2000))
Glycosidic triterpenes				
90	1444001-94-8	FF	Cytotoxic (n.a.)	Zang et al. (2013)
91	Fomitoside A	FP	Anti-inflammatory	Yoshikawa et al. (2005)
92	Fomitoside E	FP	Anti-inflammatory	Yoshikawa et al. (2005)
93	Fomitoside B	FP	Anti-inflammatory	Yoshikawa et al. (2005)
94	Fomitoside I	FP	Anti-inflammatory	Yoshikawa et al. (2005)
95	Fomitoside C	FP	Anti-inflammatory	Yoshikawa et al. (2005)
96	Fomitoside F	FP	Anti-inflammatory	Yoshikawa et al. (2005)
97	Fomitoside J	FP	Anti-inflammatory	Yoshikawa et al. (2005)
98	Fomitoside D	FP	Anti-inflammatory	Yoshikawa et al. (2005)
99	Fomitoside G	FP	Anti-inflammatory	Yoshikawa et al. (2005)
100	Fomitoside H	FP	Anti-inflammatory	Yoshikawa et al. (2005)
101	Tuberoside	FF	Cytotoxic (m.a.)	Zang et al. (2013)
Miscellaneous triterpenes				
102	(22E)-Ergosta-3,5,7,9(11),22-pentaene	LS	n.g.	Coy and Nieto (2009)
(B) Organic acids and related compounds				
103	Malonic acid	LS	n.g.	Olennikov et al. (2008)
104	Malic acid	LS	n.g.	Olennikov et al. (2008)
105	Succinic acid	LS	n.g.	Olennikov et al. (2008)
106	Tartaric acid	LS	n.g.	Olennikov et al. (2008)
107	Citric acid	LS	n.g.	Olennikov et al. (2008)
108	Masutakic acid A	LS	n.g.	Yoshikawa et al. (2001)
109	Agaric acid	LO	n.g.	Airapetova et al. (2010)

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Table 2 (*continued*)

No.	Compound name	Polypore species	Biological activity	Reference(s)
110	p-Hydroxybenzoic acid	FP, LS, PB	Antioxidant	Sulkowska-Ziaja et al. (2012)
111	Protocatechuic acid	FP, LS, PB	Antioxidant	Sulkowska-Ziaja et al. (2012)
112	Vanillic acid	FP, LS, PB	Antioxidant	Sulkowska-Ziaja et al. (2012)
113	Protocatechualdehyde	FF	n.g.	Feng and Yang (2010)
114	Gallic acid	LS	Antioxidant	Olennikov et al. (2011)
115	p-Coumaric acid	LS	Antioxidant	Olennikov et al. (2011)
116	Caffeic acid	LS	Antioxidant	Olennikov et al. (2011)
117	(3E)-4-(3,4-dihydroxyphenyl)-3-Buten-2-one	FF	n.g.	Feng and Yang (2010)
118	2-[(2E)-4-hydroxy-3-methyl-2-buten-1-yl]-1,4-Benzenediol	PB	Matrix metallo-proteinase inhibitor	Kawagishi et al. (2002)
119	Chlorogenic acid	LS	Antioxidant	Olennikov et al. (2011)
120	Pinicolic acid C	FP	n.g.	Rösecke and König (1999)
121	Laricinolic acid	LO	n.g.	Erb et al. (2000) and Wu et al. (2009)
122	Officinalic acid	LO	n.g.	Epstein et al., (1979), Erb et al., (2000) and Wu et al. (2009)
(C) Benzofurans				
123	Paulownin	FF	n.g.	Feng and Yang (2010)
124	Demethoxyegonol	LS	n.g.	Yoshikawa et al. (2001)
125	Egonol	LS	n.g.	Yoshikawa et al. (2001)
126	Egonolglucoside	LS	n.g.	Yoshikawa et al. (2001)
127	Masutakeside I	LS	n.g.	Yoshikawa et al. (2001)
128	Egonolgentiobioside	LS	n.g.	Yoshikawa et al. (2001)
129	4-[2,3-dihydro-7-hydroxy-3-(hydroxymethyl)-5-(3-hydroxypropyl)-2-benzofuranyl]-1,2-Benzenediol	LS	n.g.	Yoshikawa et al. (2001)
130	(±)-Laetirobin	LS	Cytostatic	Lear et al. (2009)
(D) Flavonoids and related compounds				
131	Kaempferol	LS	Antioxidant	Olennikov et al. (2011)
132	Quercetin	LS	Antioxidant	Olennikov et al. (2011)
133	(2R,3S)-(+)-Catechin	LS	Antioxidant	Olennikov et al. (2011)
Q13 **(E) Coumarins**				
134	Daphnetin	FF	Antitumor	Huang et al. (2012)
135	2H-6-chloro-2-oxo-4-phenyl-1-Benzopyran-3-carboxylic acid ethyl ester	LO	Antimicrobial	Hwang et al. (2013)
136	6-Chloro-4-phenyl-coumarin	LO	Antimicrobial	Hwang et al. (2013)
(F) N-containing compounds				
137	N-phenethyl-Hexadecanamide	LS	n.g.	Shiono et al. (2005)
138	Piptamine	PB	Antibiotic	Schlegel et al. (2000)

LS, *Laetiporus sulphureus*; PB, *Piptoporus betulinus*; FP, *Fomitopsis pinicola*; FF, *Fomes fomentarius*; LO, *Laricifomes officinalis*; n.g., *not given*; n.a., *no activity*; w.a., *weak activity*; m.a., *moderate activity*

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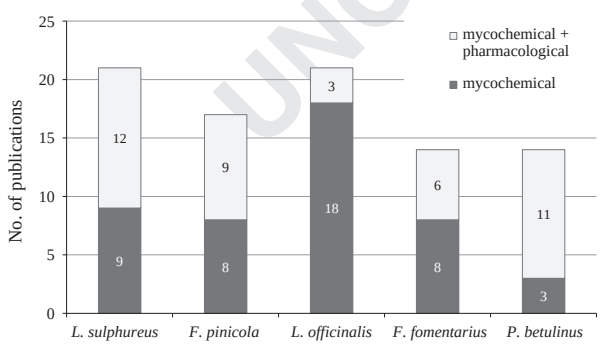

Fig. 1. Evaluation of number of publications dealing with secondary metabolites (n=87) differentiated in studies reporting mycochemical work only (n=46) and studies reporting a combination of mycochemical work and pharmacological activity (n=41).

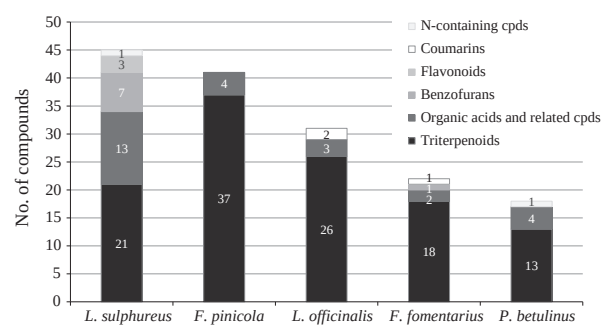

Fig. 2. Categorization of pure compoundsisolated from five selected polypore species in six chemical classes, *i.e.* triterpenes, organic acids and related compounds, benzofurans, flavonoids, coumarins, and N-containing compounds.

Reuters – http://www.webofknowledge.com) resulting in 87 publications of interest. In total more than 135 pure compounds have been isolated and identified from the five polypore species of interest (Table 2). In general, investigations carried out on *Laetiporus sulphureus*, *Fomitopsis pinicola*, and *Fomes fomentarius* show an almost equal distribution of mycochemical studies and studies combined with an evaluation of biological activities of either extracts or pure constituents (Fig. 1). However, this tendency could not be observed for *Laricifomes officinalis* and *Piptoporus betulinus*, where there is either a higher number of mycochemical reports or a higher number of combined mycochemical/pharmacological studies, respectively.

5.2. Overview on the chemical nature of secondary metabolites

Isolated and identified small molecules from the selected polypores, *i.e. Laetiporus sulphureus*, *Piptoporus betulinus*, *Laricifomes*

officinalis, *Fomitopsis pinicola*, and *Fomes fomentarius* can be classified as triterpenoids, organic acids and related compounds, benzofurans, flavonoids, coumarins, and N-containing constituents (Fig. 2).

Concerning the multiplicity of small molecule classes, high diversity is observed for *Laetiporus sulphureus* and *Fomes fomentarius*, whilst *Laricifomes officinalis*, *Piptoporus betulinus*as well as *Fomitopsis pinicola* show less diversity in chemical classes. Interestingly, only about 11% of all identified secondary metabolites have been reported as constituents of two or more of the focussed polypore species.

To extract, enrich, and isolate pure compounds, usually standard organic solvent extraction (EtOH or MeOH) and liquid-liquid partition is followed by classical column chromatography (silica gel and Sephadex LH-20). As a final step in the isolation process (semi)preparative techniques like HPLC or TLC are applied. Interestingly, more sophisticated techniques such as high-speed counter-current chromatography have been scarcely reported so far (Zhang et al., 2013). Finally, secondary metabolites are identified by LC–MS and by interpretation of 1D and 2D NMR spectra.

5.2.1. Triterpenoids

In all five polyporespecies of interest, the major part of secondary metabolites, around 75%, is composed of triterpenoids (~100 different structures), whereas other secondary metabolite classes are produced to a lesser extent.

Triterpenes, *i.e.* mainly lanostanes,biosynthesised by the selected polypores can be subdivided into acids (Fig. 3), esters and lactones (Fig. 4), alcohols (Fig. 5), ethers and peroxides, aldehydes and ketones, glycosides, and miscellaneous triterpenes (Fig. 6). Lanostanes are triterpenes usually containing 30 carbon atoms and a characteristic tetracyclic skeleton. Together with the groups of dammaranes, tirucallanes, euphanes, and cucurbitanes, they are biosynthetically derived from lanosterol.

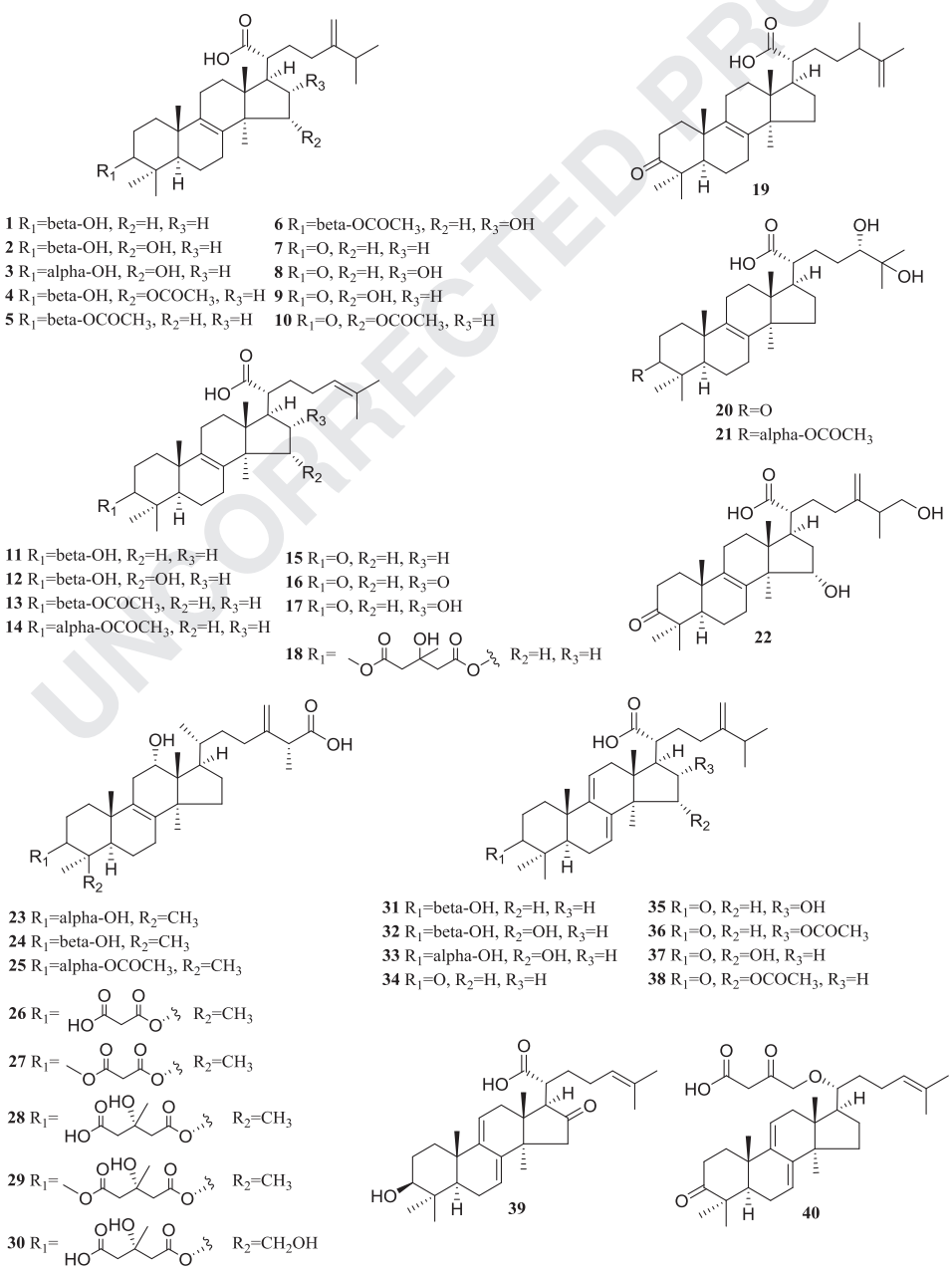

Fig. 3. Chemical structures of polypore triterpene acids.

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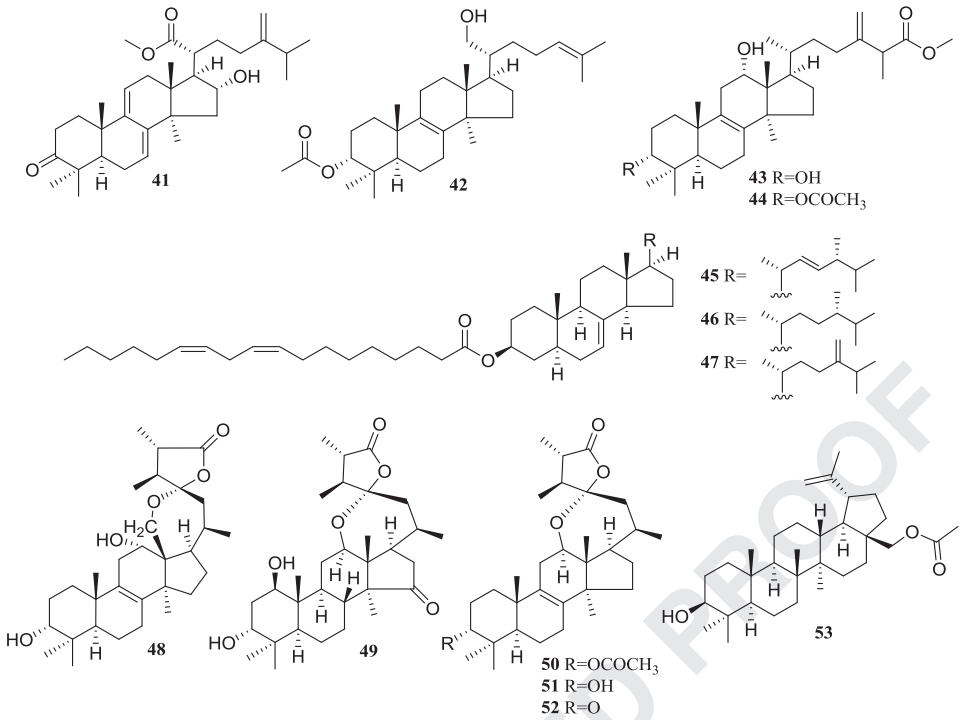

Fig. 4. Chemical structures of polypore triterpene esters and lactones.

5.2.2. Organic acids and related compounds

The second largest group of secondary metabolites (~14%) reported from the selected polypores is composed of organic acids. So far around 20 of them have been described including aliphatic, aromatic, and related compounds (Fig. 7, compounds **103–122**).

5.2.3. Other secondary metabolites including volatile components

Besides the major part of around 90% of triterpenes and organic acids, the selected polypore species also contain other compounds belonging to different chemical classes, *i.e.* benzofurans (Fig. 7, compounds **123–130**), flavonoids, coumarins, and N-containing compounds (Fig. 8).

Since these polypore species grow on wood such as tree trunks or fallen logs it is questionable, if some of these minor components are genuinely produced by the fungus or rather deducted from their substrate, *i.e.* the respective host tree bark.

This hypothesis is underlined by a study on volatile compounds from *Laetiporus sulphureus*, where results showed distinct variations in their composition with host, location, and age, even if the investigated fruiting bodies were claimed to be of the same species Q2 (S.M. Wu et al., 2005).

In general, volatile components of the selected polypores can be categorized as (i) fatty acids and methyl-branched carboxylic acids, (ii) C8 compounds and benzoic volatiles, and (iii) volatile amines (List and Menssen, 1959; Rapior et al., 2000; S.M. Wu et al., 2005). So far more than 40 major volatiles were discovered and identified by methods including (HR)GC–MS and gas chromatography–olfactometry (GC–O). To date, no biological activity has been linked to these volatile components which might be due to difficulties in assaying such compounds. However, the function of volatiles from *Fomitopsis pinicola* and *Fomes fomentarius* as insect attractants has been investigated thoroughly (Fäldt et al., 1999).

5.3. Bioactive secondary metabolites

Besides mentioned primary metabolites, a large portion of reported biological activities is closely linked to secondary metabolites. These small molecules have been found to be responsible for anti-inflammatory, cytotoxic, antimicrobial, antioxidant, and anti-thrombin properties. Despite numerous ethnomycological reports on medicinal applications, mycochemical research of native European polypores is at a very early stage (Jung et al., 2011; Lee, 2005; Petrova et al., 2008), and bioactive secondary metabolites have rarely been identified so far (Muhsin et al., 2011).

5.3.1. Biological properties of extracts or multi-component mixtures

Besides the more costly and laborious isolation of pure compounds, some research groups focus on the evaluation of the bioactive potential of extracts or multi-component mixtures. Since this chapter deals with secondary metabolites, herein discussed extracts were obtained by extraction with organic solvents (primarily EtOH or MeOH, Table 3). Organic solvent extracts from *Laetiporus sulphureus*, *Piptoporus betulinus*, *Fomitopsis pinicola*, *Fomes fomentarius* have been investigated in terms of bioactivity, whereas to the best of our knowledge, there are no reports about bioactive organic solvent extracts from *Laricifomes officinalis*. Mainly phenotypic assays were used to evaluate the extracts' biological properties, *e.g.* antioxidant, antimicrobial, and cytotoxic effects (Keller et al., 2002; Ozen et al., 2011). Only in few cases, target-specific assays have been used, *e.g.* to evaluate acetylcholinesterase (AChE) inhibitory activities of *Laetiporus sulphurous* (Orhan and Üstün, 2011).

5.3.2. Biological properties of pure compounds

The chemical structures of secondary metabolites isolated and reported to date as well as their reported biological activity are given in Table 2.

In order to obtain an unbiased overview on bioactivity, data published in μg/mL has been converted to μM and is given in brackets.

As shown in Fig. 9, the chemical class of triterpenoids might be considered as a hot spot for an abundance of biological activities comprising anti-inflammatory, cytotoxic, antimicrobial, and anti-thrombin properties. As mentioned before, around 100 triterpenoids have been reported as constituents of the five selected polypore species. Interestingly, only around 45 of these triterpenoids are reported to be biologically active.

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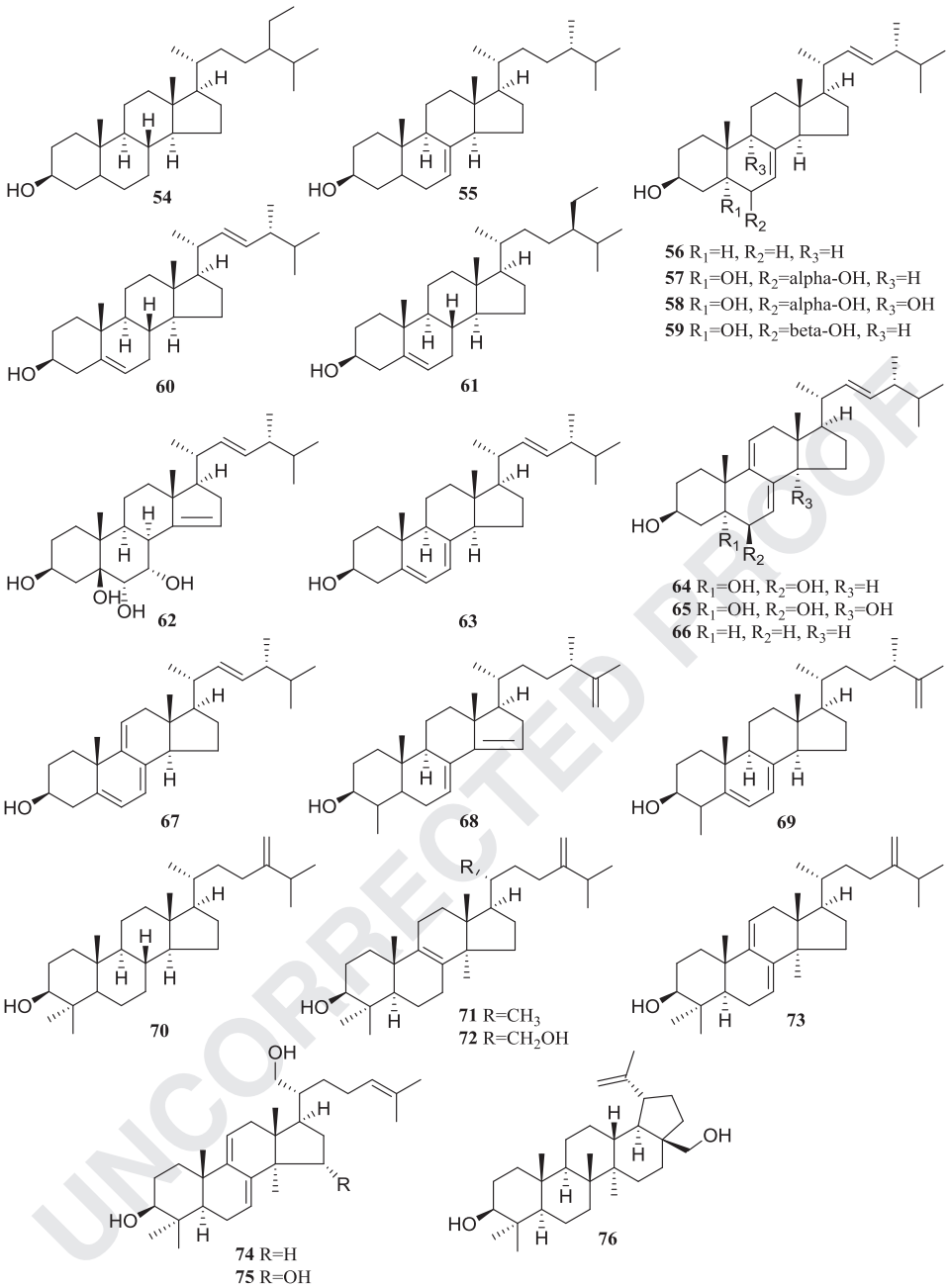

Fig. 5. Chemical structures of polypore triterpene alcohols.

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Furthermore, Fig. 9 reveals that other compound classes show less diversity in biological activities. For organic acids and related compounds for instance, mainly antioxidant properties are reported in the literature.

5.3.2.1. Anti-inflammatory effects. Several lanostane-type triterpene acids from *Piptoporus betulinus* were evaluated with respect to their anti-inflammatory activity. The triterpene acids **25**, **27**, **29**, and **35** showed weak cyclooxygenase-1 (COX-1) inhibition. However, a distinct inhibitory activity was found against 3α-hydroxysteroid dehydrogenase (3α-HSD), a key enzyme in androgen metabolism. Moreover, strong selective bacterial hyaluronidase inhibition was determined for these four lanostanes (Wangun et al., 2004).

Besides before-mentioned compounds **29** and **35**, also the triterpene acids **23**, **26**, **28**, and **30** were investigated in an *in vivo* model on their ability to suppress 12-O-tetradecanoylphorbol-13-acetate (TPA)-induced mouse ear oedema. All six tested compounds showed a promising activity by suppressing the oedema at a percentage between 49% and 86% at a concentration of 400 nmol/ear (Kamo et al., 2003).

Together with ten glycosidic triterpenes (fomitoside A–J; **91–100**), two lanostane acids (fomitopinic acid A and B, **20** and **21**) from *Fomitopsis pinicola* were investigated for their COX-1 and COX-2 inhibitory activity. Especially, compounds **20**, **92**, and **96** showed a potent and selective *in vitro* inhibition of COX-2 (IC_{50} values 0.15–1.15 μM; pos. ctrl.: indomethacin, IC_{50} 0.60 μM; aspirin, IC_{50} 4.97 μM) (Yoshikawa et al., 2005). IC_{50} values for **21**, **91**, **93**, **94**, and **97**are not reported.

5.3.2.2. Anti-cancer, anti-tumour, and cytotoxic effects. Compounds discussed in this chapter might be referred to as "anti-tumour" or "anti-cancer" in the original literature. In the present review article, these terms are only used for compounds which inhibit

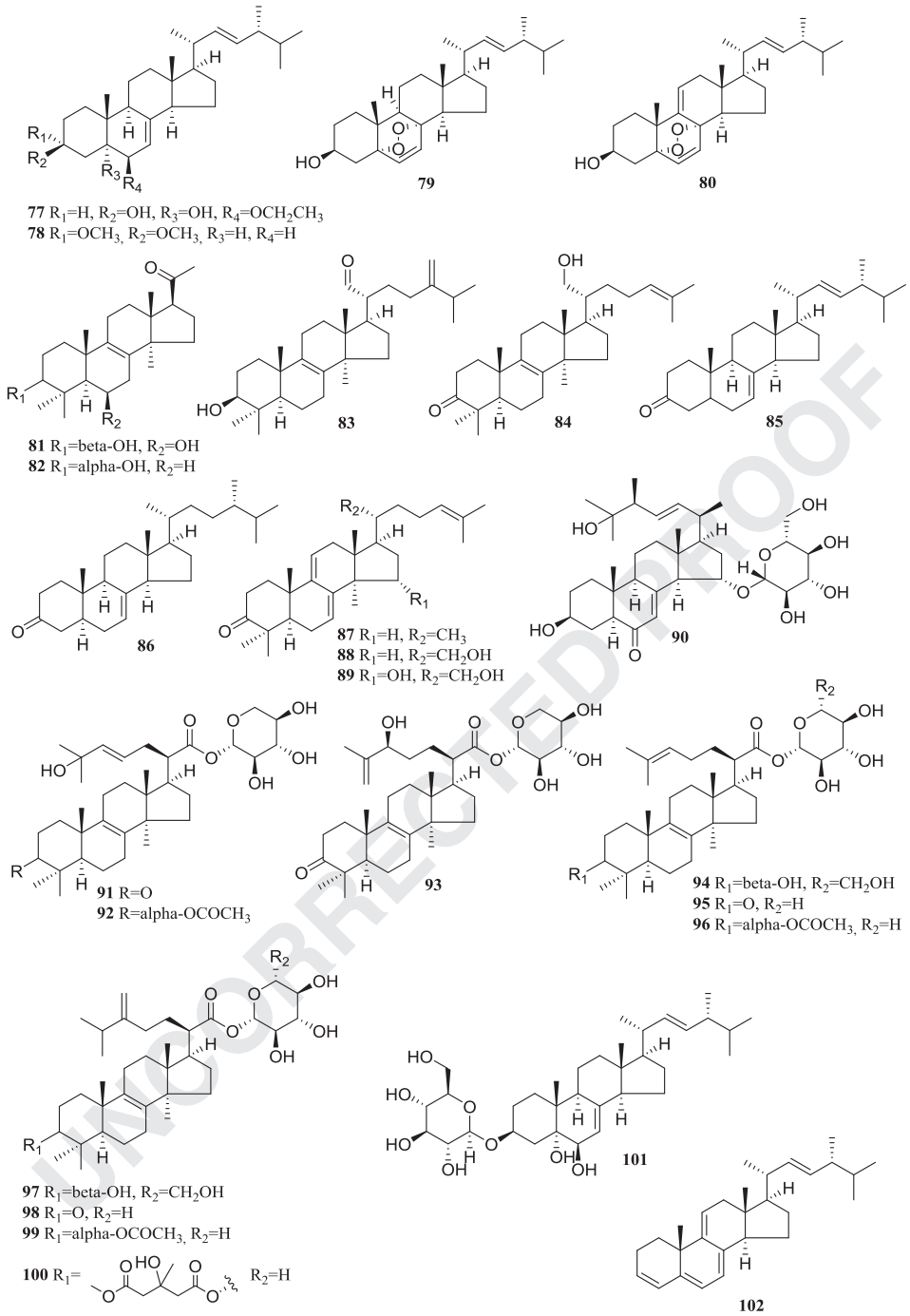

Fig. 6. Chemical structures of polypore triterpene ethers and peroxides, aldehydes and ketones, glycosides, and miscellaneous triterpenes.

the growth of tumours in animal-based models or which show distinct activity in human-based clinical studies. In most cases, studies on polypore constituents deal with compounds which suppress the growth of or kill isolated tumour cell lines; hence these are referred to as either cytostatic or cytotoxic, respectively.

Six lanostane-type triterpene acids from *Laetiporus sulphureus*, *i.e.* eburicoic acid (**1**), sulfurenic acid (**2**), 15α-hydroxytrametenolic acid (**12**), 3-O-acetyleburicoic acid (**5**), (3β)-3-(acetyloxy)-lanosta-8,24-dien-21-oic acid (**13**), and fomefficinic acid D (**9**),and the semi-synthetic compound versisponic acid C (**10**) were assayed in an MTT assay (pos. ctrl.: ursolic acid, IC_{50} 21 μM) for their apoptotic potential against HL-60 cells (human myeloid leukaemia cells) (León et al., 2004). Most promising dose-dependent inhibition of the proliferation of these cancer cells was observed for **2**, **5**, and **12**, showing IC_{50} values of 14, 15, and 12 μM, respectively. Compounds **1**, **13**, and **10** showed moderate activity with IC_{50} values in the range of 25 to 31 μM, whereas **9** gave only a very weak activity with an IC_{50} value of 407 μM. Furthermore, this research group performed a quantitative fluorescence microscopy study on key proteins for proteolytic cleavage in the process of apoptosis. Hereby, compound **5** was identified as most promising inducer of PARP-1 cleavage (pos. ctrl.: etoposide). The triterpene acids **5** and **10** also showed a positive effect on the release of cytochrome c from mitochondria into the cytosol (León et al., 2004). By relating these results to structural features one may conclude that acetylated triterpenes show more potent effects in the mentioned mechanisms than non-acetylated derivatives (León et al., 2004).

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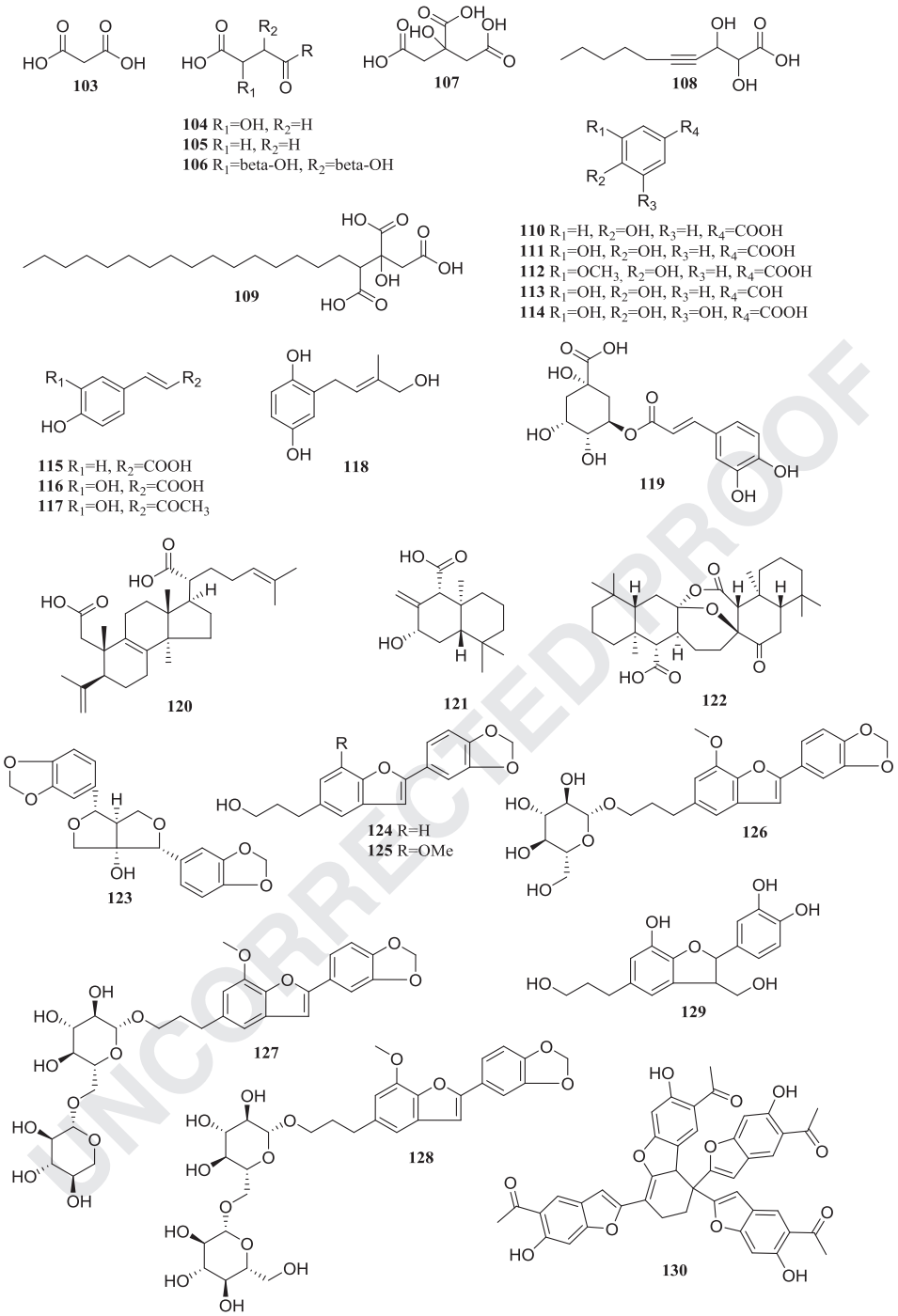

Fig. 7. Chemical structures of polypore organic acids, related structures, and benzofurans.

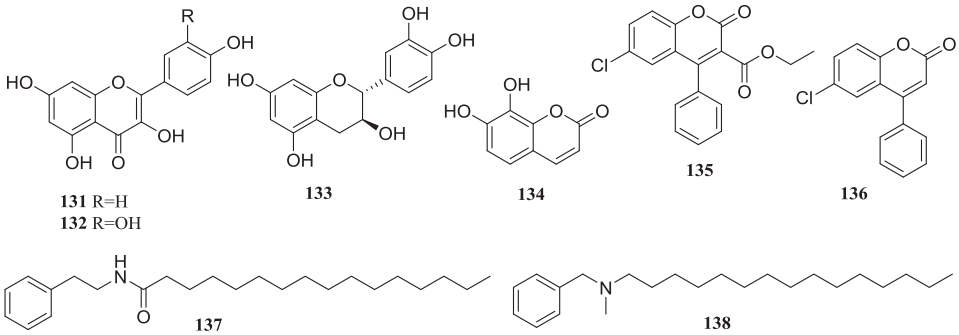

Fig. 8. Chemical structures of polypore flavonoids (131–133), coumarins (134–136), and N-containing compounds (137 and 138).

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Table 3
Overview on bioactive polypore extracts obtained by extraction with organic solvents.

Species	Type of extract	Biological activity	Reference(s)
LS	MeOH	Antioxidant, antimicrobial	Ozen et al. (2011)
LS, PB	MeOH 70%; chloroform	Antimicrobial	Karaman et al. (2009)
LS	n.g.	Antimicrobial	Demir and Yamac (2008)
LS	EtOH 85%	Antioxidant, AChE inhibition	Orhan and Üstün (2011)
LS	MeOH 70%	Antioxidant, antimicrobial	Karaman et al. (2010)
LS	EtOH	Antioxidant, antimicrobial	Turkoglu et al. (2007)
LS	DCM; MeOH	Antibacterial	Keller et al. (2002)
PB	n.g.	Cytotoxic	Lemieszek et al. (2009)
PB	ether; EtOH	Antiproliferative against cancer-derived cells	Cyranka et al. (2011)
PB	Ether	Interference inducing	Kandefer-Szerszeń and Kawecki (1974)
PB	n.g.	Inhibiting growth of mouse sarcoma S-37	Blumenberg and Kessler (1963)
FP	EtOH	Antioxidant	Macakova et al. (2010)
FP	EtOH	Anti-inflammatory	Cheng et al. (2008)
FP	EtOH 95%; MeOH	Antioxidant, cytotoxic	Choi et al. (2007)
FP	Petroleum ether; EtOAc; MeOH	Cytotoxic	Ren et al. (2006)
FP	Chloroform; n-BuOH	Antimicrobial	Petrova et al. (2007)
FP	Chloroform; EtOH	Antifungal	Guler et al. (2009)
FF	MeOH	Anti-inflammatory, anti-nociceptive	Park et al. (2004)

LS, *Laetiporus sulphureus*; PB, *Piptoporus betulinus*; FP, *Fomitopsis pinicola*; FF, *Fomes fomentarius*; n.g., *not given*

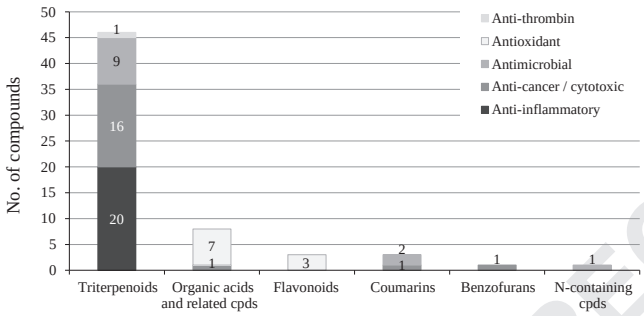

Fig. 9. Evaluation of number of bioactive secondary metabolites isolated from the five selected polypore species within respective chemical class. Reported bioactive properties are differentiated in various shades of grey.

In the compound class of benzofurans, (±)-laetirobin (**130**) from *Laetiporus sulphureus* was identified as cytostatic compound with rapid cell entry (Lear et al., 2009). The compound blocked cell division at a late stage of mitosis and invoked apoptosis. Interestingly, the investigated fungus grew as a parasite on *Robinia pseudoacacia*, the black locust tree, which suggests that compound **130** might have been produced by the host tree rather than the fungus from which it was isolated (Lear et al., 2009).

The frequently occurring mushroom constituent ergosterol peroxide (**79**) isolated from *Laetiporus sulphureus* (Krzyczkowski et al., 2009) revealed significant cytotoxicity against a human gastric cancer cell line (SNU-1, IC_{50} 18.7 μM), a human hepatoma cell line (SNU-354, IC_{50} 158.2 μM), and weak cytotoxic activities against a human colorectal cancer cell line (SNU-C4, IC_{50} 84.6 μM), and murine sarcoma-180 (IC_{50} 74.1 μM) (Nam et al., 2001).

Huang and co-workers investigated constituents of *Fomes fomentarius* for their cytotoxic activity where the triterpenes (22*E*)-ergosta-7,22-dien-3-one (**85**) and (+)-betulin (**76**) showed the strongest effects against NCI-H 460 and SGC-7901 cells, respectively (Huang et al., 2012).

In another study dealing with compounds isolated from *Fomes fomentarius*, cytotoxic effects against several human cancer cell lines were tested either in a standard MTT based colorimetric assay (against HCT116 and H1299) or in a CellTiterGlo™ luminescent cell viability assay (against A549, MCF-7, NUGC-3, SHSY-5Y, SNU739). The glycosidic triterpenoid tuberoside (**101**) showed moderate cytotoxic effects against lung (A549), breast (MCF-7), and gastric (NUGC-3) carcinoma cells with IC_{50} values between 24 and 30 μM. Only weak cytotoxicity (IC_{50} 125–180 μM) against colon (HCT116) cancer cells was determined for **58** and two novel triterpenes (**62** and **77**) (Zang et al., 2013).

As reported by the group of Kawagishi and co-workers, the hydroquinone **118** and the triterpene acid **35** isolated from *Piptoporus betulinus* were identified as matrix metallo-proteinase (MMP) inhibitors (Kawagishi et al., 2002).Compound **118** was active against MMP-1 (IC_{50} 28 μM), MMP-3 (IC_{50} 23 μM), and MMP-9 (IC_{50} 37 μM) whereas **35** showed only a moderate inhibitory activity against MMP-1 (IC_{50} 126 μM) (Kawagishi et al., 2002).

5.3.2.3. Antimicrobial effects. Recent studies on the anti-tuberculosis potential of chlorinated coumarins, *i.e.* 6-chloro-4-phenyl-coumarin (**136**) and 2H-6-chloro-2-oxo-4-phenyl-1-benzopyran-3-carboxylic acid ethyl ester (**135**), isolated from *Laricifomes officinalis* and two corresponding structural congeners revealed that coumarins containing an ethyl ester in position 3 and chlorine in position 6 show higher activity than derivatives with chlorine in position 7. However, the 7-chloro congener of **136** showed the highest activity against both replicating and non-replicating *Mycobacterium tuberculosis* (MICs of 23.9 μg/mL (=93.1 μM) and 21.9 μg/mL (=85.3 μM)) (Hwang et al., 2013).

Further investigations on the antimicrobial activity of chlorinated coumarins (c=100 μg/mL) resulted in no activity against several Gram-positive (*Staphylococcus aureus*, *Enterococcus faecalis*, *Streptococcus pneumonia*) and Gram-negative (*Escherichia coli*, *Pseudomonas*, *Acinetobacter baumanii*) bacteria, as well as against the mycobacterium *Mycobacterium smegmatis* and the fungus *Candida albicans* (Hwang et al., 2013). However, tests for antimicrobial activity against several non-tuberculous mycobacteria including *Mycobacterium chelonae*, *Mycobacterium abscessus*, *Mycobacterium marinum*, *Mycobacterium kansasii*, *Mycobacterium avium*, and *Mycobacterium bovis* resulted in an MIC of 97.1 μg/mL (=295.4 μM) against *Mycobacterium marinum* for **135** and in an MIC of 49.3 μg/mL (=192.1 μM) against *Mycobacterium kansasii* for the 7-chloro congener of **136** (Hwang et al., 2013). Q3

Further antimicrobial substances have been identified from the structure class of triterpenes. For example, five lanostanoid derivatives, *i.e.* (+)-trametenolic acid B (**11**), pachymic acid (**6**), tsugaric acid A (**14**), fomitopsic acid (**18**), pinicolic acid A (**15**), **35**, α-dihydroergosterol (**56**), and (+)-21-hydroxylanosta-8,24-dien-3-one (**84**), isolated from the polypore *Fomitopsis pinicola* were evaluated for their antimicrobial activity against *B. subtilis* in a TLC-based bioassay. Except for **6**, **56**, and **84**, all triterpenes

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showed antimicrobial activity in the range of 0.01–1 μg (pos. ctrl.: chloramphenicol, active at 0.01 g) (Keller et al., 1996). However, in a classical agar dilution assay up to a concentration of 50 μg/mL there was no inhibitory activity found against *B. subtilis* (Keller et al., 1996).

Liu and co-workers tested further triterpenoids from *Fomitopsis pinicola* for their activity against *Bacillus cereus* using a standard disc diffusion assay (Liu et al., 2010).Compound **15** which was active against *B. subtilis* was also tested in this study. Moreover, 24-methyl-3-oxo-lanosta-8,25-dien-21-oic acid (**19**), 16α-hydroxyeburiconic acid (**8**), ergosterol D (**66**), **35**, and **36** were evaluated for their antimicrobial activity against *Bacillus cereus*. In parallel, the cytotoxicity of the compounds has been assessed in order to distinguish between unspecific and specific antibacterial effects (Liu et al., 2010). As a result, **19**, **15**, **35**, and **66** were proposed to be distinctly antibacterial since their antimicrobial MICs are 15- to 30-fold lower than their respective cytotoxic IC_{50}'s (Liu et al., 2010).

The N-containing compound piptamine (**138**)isolated from *Piptoporus betulinus* was tested against a panel of Gram-positive bacteria, yeasts, and fungi by using an agar diffusion assay or by applying a standard antimicrobial susceptibility test (for aerobically growing bacteria) or in the case of yeast by using a broth dilution antifungal susceptibility test (Schlegel et al., 2000). For compound **138**, the most promising MIC values of 0.78 μg/mL (=2.35 μM) and 1.56 μg/mL (=4.70 μM) were obtained against *Staphylococcus aureus* and *Enterococcus faecalis*, respectively. Furthermore, **138** was tested for its haemolytic activity which was determined to be at 10–50 μg/mL (=30–150 μM) using heparinized blood of Beagle dogs (Schlegel et al., 2000).

5.3.2.4. Antioxidant effects. As listed in Table 3, many research groups focus on the evaluation of antioxidant effects of polypore extracts. However, only few studies aim at the isolation of pure constituents and the identification of the bioactive principle behind these effects. In general, antioxidant properties of discussed polypore species refer to flavonoids such as kaempferol (**131**), quercetin (**132**), and (2R,3S)-(+)-catechin (**133**) and organic acids such as gallic acid (**114**), p-coumaric acid (**115**), caffeic acid (**116**), and chlorogenic acid (**119**) (Olennikov et al., 2011). These well-known compounds can be considered as almost omnipresent in natural products and are not specific for the mentioned polypore species. Moreover, their quantity in the fruit body materials as well as the impact of their antioxidant and radical-scavenging properties is considered as rather low.

5.3.2.5. Other effects. The triterpene acid versisponic acid D (**4**) isolated from the chloroform extract of *Laricifomes officinalis* distinctly inhibited thrombin. However, further tested triterpene acids from this fungus, including**1**, **2**, dehydroeburicoic acid (**31**), dehydrosulfurenic acid (**32**), dehydroeburiconic acid (**34**), and 3-ketodehydrosulfurenic acid (**37**), did not show any anti-
Q4 thrombin activity (X. Wu et al., 2005).

6. A modern view on traditional uses

The use and application of polypores as commodities, food, or medicine is a cultural issue. About 65 years ago, Wasson and Wasson first described this phenomenon and reported a striking difference between mycophilic and mycophobic people (Hawksworth, 1996; Wasson and Wasson, 1957); differences, which are based on family of languages and the cultural exchange among people in Europe (Peintner et al., 2013). However, this does not explain why the mycophilic Italians or French do not extensively use medicinal poylpores up to the present. It was regarded as old-fashioned to use natural remedies, when people were wealthy enough and had access to modern drugs. But the most important factor for the strikingly rare use of medicinal fungi in Central European folk medicine is the influence of the catholic church, directly connecting fungi with devil and witchcraft: *e.g.* the medieval fresco from Plaincourault (1291) is depicting a serpent passing the "fruit of the mushroom tree" to Adam (Molitoris, 2005). The knowledge around and the use of fungi was discredited and fungi were banned from the world of good spirits and displaced into the world of devil and superstition (Kreisel, 1997; Marzell, 1921; Wasson, 1969). Thus, we deduce that medicinal polypores have been widely used in the whole Eurasian area, but Central Europeans have strongly reduced using medicinal polypores, probably in early medieval times.

6.1. Traditional uses of polypores: myths, religion, and medicine

Native people have always been connecting physical illness with bewitchment or supernatural forces. Therefore, for curing illness they often combined special rites, *e.g.* fumigation, devotions or art, with the application of traditional medicinal plants or fungi. Psychoactive mushrooms have been especially important in myth and religion, the ancient Soma cult being one prominent example: it involved shamanic experience induced by the drinking of a sacred drug, a fungal extract of *Amanita muscaria*. This was regarded as "elixir of eternal happiness" in the thousand-year-old Indian Rig Vedas. The fly agaric was also important in the Christian religion, as it forms the mushroom tree, shown on the medieval frescos of a church in Plaincorault (1291). In popular belief, this fungus is still considered to bring luck and happiness (Molitoris, 2005). In 1964, Takemoto and colleagues identified ibotenic acid and muscimol as the psychoactive compounds responsible for the effects, directly connecting an ethnomycological application with bioactive compounds (Takemoto et al., 1964).

Medicinal polypores were often regarded as representing eternal strength and including spiritual power. Therefore, they were also used as raw material for cultic art objects. The best known examples are carvings made of the "Bread of the Ghosts", which were fruiting bodies of *Laricifomes officinalis*. The shape of the carved fruiting bodies often included mouth and/or stomach orifices, which gave the mushroom spirit-catching abilities. The use of *Laricifomes officinalis* is of particular interest because it was crafted and imparted spiritualistic power through shape and substance (Blanchette et al., 1992; Nicholson, 2009). Polypore carvings or ornamented fruiting bodies are also known for *Piptoporus betulinus* (Lohwag, 1965; Peintner et al., 1998) and *Haploporus odorus* (Blanchette, 1997).

6.2. From ethnomycological application to bioactive metabolites

Biochemical and pharmacological research on remedies used in traditional medicine aims to clarify the empirical basis of the medicinal properties of plants and fungi, which were passed down from generation to generation. *Claviceps purpurea* and the discovery of ergot is another prominent example for an application starting from witchcraft to modern biotechnology (Haarmann et al., 2009).

For European medicinal polypore species, research is just beginning, but promising results have already been obtained for all five polypore species included in this review: the traditional application of *Laricifomes officinalis* in folk medicine against fever and sweat related to tuberculosis was underlined by a recent study identifying two new chlorinated coumarins as bioactive compounds effective against *Mycobacterium tuberculosis* (Hwang et al., 2013). This finding corroborates the early hypothesis that the fungus called agarikon (αγαρικόν) which was used against tuberculosis is *Laricifomes officinalis*.

Fomes fomentarius was widely used throughout Europe as a styptic and as absorbing wound bandage. Cytotoxic and

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anti-tumour effects have been reported for fruit body extracts of this polypore confirming the effect of this medicinal polypore in the treatment of cancer; but antimicrobial properties have not yet been extensively tested (Table 2). On the other hand modern research helped to detect antibiotic and anti-inflammatory effects of *Piptoporus betulinus* constituents, which was also used as a styptic or antiseptic. The application of *Fomitopsis pinicola* is not clearly passed down, but modern studies showed that this polypore has anti-inflammatory, anti-microbial and anti-oxidant properties (Table 2). Many unspecific positive effects of medicinal polypores might be related to immuno-enhancing properties of fungal polysaccharides: *Laetiporus sulphureus* has been applied for improving health and defending the body against illnesses; the fruit bodies are a rich source for glucans and polysaccharides which activate immune-modulating mediators, and provoke a hypoglycaemia effect (Table 2).

7. Potential and challenges of polypores in mycochemistry and modern medicine

In the past few decades, many mushrooms have been used as a valuable source for bioactive compounds, for therapeutic adjuvants, or as health promoting food supplements. However, it is very difficult, often even impossible, to compare and evaluate results from different studies focussing on the effects shown by certain medicinal polypores due to the following three major pitfalls: (i) insufficiently characterized fungal starting material, (ii) varying extraction methods, and (iii) different biological test set-ups. Hence, this chapter aims at discussing the most important aspects, which might be worth to consider when working with medicinal polypores.

7.1. Origin of the fungal material

Studies focussing on chemical compounds from fungi and their biological effects are either based on fruit body material collected in the wilderness or from cultivated material. Alternatively, mycelial cultures obtained from fruit bodies can also serve as starting material for mycochemical processing. Concerning polypores collected from the wild, it is essential to document from which habitat and substrate the fruit body was taken, and to deposit a voucher specimen in a public mycological collection (Agerer et al., 2000), whilst fungal cultures should be kept in a dedicated culture collection. This allows for checking or re-identifying the fungal material. Meanwhile, it has become common knowledge that polypores are not considered as plant material but that they belong to the separate regnum of fungi (Bruns, 2006).

7.2. Species delimitation and identification

Mycology is a comparatively young discipline and fungal taxonomy is still in a constant state of flux. This also applies to widely recognized polypore species such as *Fomes fomentarius* or *Laetiporus sulphureus* (Banik et al., 2012; Binder et al., 2013; McCormick et al., 2013). Many polypore species have been shown to rather represent a species complex. These complexes include distinct species with morphological, ecological, geographical, or substrate-related characters, as well as genetically divergent, cryptic species. Wrong identification of the raw material makes a correct interpretation of results and a comparison with other studies impossible. Therefore, the knowledge and support of professional mycologists is essential for an accurate identification (Fischbein et al., 2003). Moreover, reliable species identification must be based on both, classical morphological–ecological characteristics and molecular data. Interpretation of microscopic features, the existence of morphologically similar species, and the specious nature of the kingdom fungi are perpetual challenges for mycologists and make DNA barcoding essential. In this respect, the rDNA ITS region has now been widely accepted as a barcoding region for fungi (Schoch et al., 2012). The DNA sequences of the utilized fungi should be submitted to a public database when publishing results.

7.3. Fungal nomenclature

Fungal nomenclature follows the "International Code of Nomenclature for algae, fungi, and plants" (http://www.iapt-taxon.org/nomen/main.php) and is also in permanent flux; the same fungal taxon may have been described in several countries, in different languages, or from different substrates. This is the reason why most fungal taxa have several legitimate names (*e.g. Fomitopsis marginata*) but only one current name (*e.g. Fomitopsis pinicola*). Fungal names can be searched online in MycoBank (http://www.mycobank.org/DefaultInfo.aspx?Page=Home) or in the Index Fungorum (http://www.indexfungorum.org/names/Names.asp). The knowledge of both the current name and older synonyms allows for comparison with older studies relying on an old name for the same species (*e.g. Laricifomes officinalis = Polyporus officinalis = Faricifomes officinalis*).

7.4. Metabolite production depending on fungal strains and substrate

Different fungal strains can exert different bioactivities depending on the genetic equipage, the geographical provenance, and the substrate, thus showing different metabolite profiles and different functions (Agafonova et al., 2007; Meng et al., 2013; Szedlay et al., 1999). Metabolite production and function also differ depending on the type of fungal material used: fruit bodies, mycelial cultures, or culture filtrates (Cui and Chisti, 2003; Erkel and Anke, 2008; Peng et al., 2005). The best-known example for this is pleuromutilin, which is derived from mycelial cultures of the genus *Clitopilus* (Hartley et al., 2009) but not from its fruit bodies. For reliable results it is therefore essential to work with fungal isolates, which allow for a perpetual production of the fungal material with the highest metabolite production (*e.g.* cultivated fruit bodies or mycelial culture) to compare metabolite production from both, fruit bodies and submerged cultures, and to deposit the fungal strain in a culture collection.

7.5. The effect of culture conditions

Traditional folk medicine uses wild growing polypore fruit bodies. A point to consider is that polypores degrade their substrate and may transform compounds from the host that cannot be produced by the fungal enzyme machinery (Paterson, 2006). This may be true particularly for lignin-derived compounds, as white rot fungi degrade lignin to monomeric phenols and further to CO_2 and H_2O for production of energy (Leonowicz et al., 1999). Q5

Primary and secondary metabolites, which are of pharmaceutical interest are usually produced in different amounts, depending on the cultivation conditions; thus, studying the cultivation conditions *in vitro* for optimization of their production is of major importance (Vieira et al., 2008). The parameters which trigger the production of bioactive compounds (especially of secondary metabolites) are still widely unexplored in European medicinal polypores: generally, the main parameters which affect the production and concentration of bioactive compounds in polypores are fungal strain, substrate, pH, temperature, but also the addition of precursors to the culture medium may facilitate or induce

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Q6 bioactive metabolite production (Shu et al., 2004; Zhong and Tang, 2004). Submerged culture of medicinal polypores has significant industrial potential. It is therefore essential to optimize and standardize the culture conditions in order to allow for reproducible and reliable results.

7.6. Crude extracts versus pure substances

Most of the pharmacological studies on bioactive compounds of polypores were conducted using crude and poorly characterized extracts. The possible mechanism of action as well as potential synergistic or antagonistic effects of multi-component mixtures derived from polypores need to be evaluated integrating pharmacological, pharmacokinetic, bioavailability-centred and physiological approaches. In addition, more experiments including *in vitro*, *in vivo* and clinical studies should be carried out in order to identify side effects or to assess potential toxicity issues.

8. Conclusion and future perspectives

There is a renewed and increasing interest in using native mushroom species as "botanicals", or better "mycologicals", in modern Western medicine. Studying the rather unexplored medicinal potential of European polypore species seems to be a promising endeavour, since therapeutic effects of fruit body extracts or isolated pure compounds have already been documented for other polypores originating for instance from Asia (Herrmann, 1962; Killermann, 1938; Peintner et al., 1998; Sporenheimer, 1936; Wasser and Weis, 1999). Moreover, the traditional use over centuries indicates that potential health risks which might be caused by these fungi can be considered as low. Besides nutrition, the consumption of medicinal mushrooms can be beneficial to humans through their ability to cure various diseases (Ferreira et al., 2010; Hobbs, 1995).

Studies on the activities and mechanisms of action of fungal metabolites of European medicinal polypores are urgently needed to develop them as modern evidence-based medicinal products. Here, the issues of dosage, bioavailability, and synergisms should not be neglected since most of the published results are basically *in vitro* studies and clinical evidence is largely missing (Gertsch, 2011). New methods and sophisticated analytical techniques integrated with biotechnology and other relevant disciplines are urgently required (De Silva et al., 2012). *in vivo* experimentation with high-quality, long-term double-blinded, clinical studies including large trial populations are essential to confirm the safety and effects of fungus-derived compounds. In order to deal with the complexity of the large number of secondary as well as primary metabolites, sophisticated modern methods including chemometrics, "omics", or systems-wide approaches should be considered to obtain insights into the metabolomic regulations within the polypore species of interest.

More than 30 species of medicinal mushrooms are currently identified as sources for biologically active metabolites. Much of the evidence is based on traditional medicine resulting from *in vitro* assays; only few conclusive *in vivo* data and/or clinical tests have been performed (Jikai, 2002; Thyagarajan-Sahu et al., 2011; Weng and Yen, 2010). *in vitro* tests in combination with phenotypic assaying may give an indication as to the potential therapeutic effect and mark the first steps in preclinical screening (De Silva et al., 2012). Therefore, future investigations should be directed towards the characterization of multi-component mixtures from unambiguously identified fungal material and constituents thereof, their biological effects including molecular mechanism and bioavailability to rationally develop them as accredited therapeutic and health promoting agents.

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References

Agafonova, S.V., Olennikov, D.N., Borovskii, G.B., Penzina, T.A., 2007. Chemical composition of fruiting bodies from two strains of *Laetiporus sulphureus*. Chemistry of Natural Compounds 43, 687–688.
Agerer, R., Ammirati, J., Blanz, P., Courtecuisse, R., Desjardin, D.E., Gams, W., Hallenberg, N., Halling, R., Hawksworth, D.L., Horak, E., Korf, R.P., Mueller, G. M., Oberwinkler, F., Rambold, G., Summerbell, R.C., Triebel, D., Watling, R., 2000. Always deposit voucher specimens – open letter to the scientific community of mycologists. Nova Hedwigia 71, 539–543.
Airapetova, A.Y., Gavrilin, M.V., Dmitriev, A.B., Mezenova, T.D., 2010. Examination of the structure of agaricinic acid using 1H and 13C NMR spectroscopy. Pharmaceutical Chemistry Journal 44, 510–513.
Anderson, C., Epstein, W., 1971. Metabolic intermediates in the biological oxidation of lanosterol to eburicoic acid. Phytochemistry 10, 2713–2717.
Anderson, C.G., Vanlear, G., Epstein, W.W., 1972. Minor triterpenes of *Fomes officinalis*. Phytochemistry 11, 2847–2852.
Appleton, R.E., Jan, J.E., Kroeger, P.D., 1988. *Laetiporus sulphureus* causing visual hallucinations and ataxia in a child. Canadian Medical Association Journal 139, 48–49.
Banik, M.T., Lindner, D.L., Ortiz-Santana, B., Lodge, D.J., 2012. A new species of *Laetiporus* from the Caribbean basin. Kurtziana 37, 15–21.
Berendes, J., 1902. Des Pedanios Dioskurides aus Anazarbos Arzneimittellehre in fünf Büchern (reprint). Sändig Reprints Verlag, Vaduz, Lichtenstein p. 1997.
Binder, M., Justo, A., Riley, R., Salamov, A., López-Giráldez, F., Sjökvist, E., Copeland, A., Foster, B., Sun, H., Larsson, E., Larsson, K.-H., Townsend, J., Grigoriev, I.V., Hibbett, D.S., 2013. Phylogenetic and phylogenomic overview of the polyporales. Mycologia 105, 1350–1373.
Blanchette, R.A., 1997. Haploporus odorus: a sacred fungus in traditional Native American culture of the northern plains. Mycologia 89, 233–240.
Blanchette, R.A., Compton, B.D., Turner, N.J., Gilbertson, R.L., 1992. Nineteenth century Shaman grave guardians are carved *Fomitopsis officinalis* sporophores. Mycologia 84, 119–124.
Blumenberg, F., Kessler, F., 1963. Inhibition of the growth of mouse sarcoma S-37 by the birch fungus (*Polyporus betulinus*). Arzneimittel-Forschung 13, 198–200.
Bruns, T., 2006. Evolutionary biology: a kingdom revised. Nature 443, 758–761.
Bryce, T.A., Campbell, I.M., McCorkindale, N.J., 1967. Metabolites of the polyporaceae—I : Novel conjugates of polyporenic acid a from *Piptoporus betulinus*. Tetrahedron 23, 3427–3434.
Buller, A., 1914. The fungus lore of the Greeks and Romans. Transactions of the British Mycological Society 5, 21–66.
Burdsall, H., Banik, M.T., 2001. The genus *Laetiporus* in North America. Harvard Papers in Botany 6, 43–55.
Cha, W.-S., Ding, J.-L., Shin, H.-J., Kim, J.-S., Kim, Y.-S., Choi, D., Lee, H.-D., Kang, H.-B., Lee, C.-W., 2009. Effect of *Fomitopsis pinicola* extract on blood glucose and lipid metabolism in diabetic rats. Korean Journal of Chemical Engineering 26, 1696–1699.
Chang, M.-Y., Tsai, G.-J., Houng, J.-Y., 2006. Optimization of the medium composition for the submerged culture of *Ganoderma lucidum* by Taguchi array design and steepest ascent method. Enzyme and Microbial Technology 38, 407–414.
Chang, S.-T., 1999. Global impact of edible and medicinal mushrooms on human welfare in the 21st century: nongreen revolution. International Journal Medicinal Mushrooms 1, 1–8.
Chen, W., Zhao, Z., Chen, S.-F., Li, Y.-Q., 2008. Optimization for the production of exopolysaccharide from *Fomes fomentarius* in submerged culture and its antitumor effect *in vitro*. Bioresource Technology 99, 3187–3194.
Chen, W., Zhao, Z., Li, Y., 2011. Simultaneous increase of mycelial biomass and intracellular polysaccharide from *Fomes fomentarius* and its biological function of gastric cancer intervention. Carbohydrate Polymers 85, 369–375.
Chen, Z., Wu, P., Xue, J., Wei, X., 2005. Triterpenes from the mycelial cultures of two *Polyporus* strains. Redai Yaredai Zhiwu Xuebao 13, 399–402.
Cheng, J.-J., Lin, C.-Y., Lur, H.-S., Chen, H.-P., Lu, M.-K., 2008. Properties and biological functions of polysaccharides and ethanolic extracts isolated from medicinal fungus *Fomitopsis pinicola*. Process Biochemistry 43, 829–834.
Choi, D., Park, S.-S., Ding, J.-L., Cha, W.-S., 2007. Effects of *Fomitopsis pinicola* extracts on antioxidant and antitumor activities. Biotechnology and Bioprocess Engineering 12, 516–524.
Comandini, O., Eros-Honti, Z., Jakucs, E., Flores Arzu, R., Leonardi, M., Rinaldi, A.C., 2012. Molecular and morpho-anatomical description of mycorrhizas of *Lactarius rimosellus* on *Quercus* sp., with ethnomycological notes on *Lactarius* in Guatemala. Mycorrhiza 22, 279–287.
Cordier, F., 1870. Les Champignons.
Coy, E.D., Nieto, I.J., 2009. Sterol composition of the macromycete fungus *Laetiporus sulphureus*. Chemistry of Natural Compounds 45, 193–196.

UNCORRECTED PROOF

Cui, J., Chisti, Y., 2003. Polysaccharopeptides of *Coriolus versicolor*: physiological activity, uses, and production. Biotechnology Advances 21, 109–122.
Cyranka, M., Grqz, M., Kaczor, J., Kandefer-Szerszen, M., Walezak, K., Kapka-Skrzypczak, L., Rzeski, W., 2011. Investigation of antiproliferative effect of ether and ethanol extracts of birch polypore medicinal mushroom, *Piptoporus betulinus* (Bull.: Fr.) P. Karst. (higher basidiomycetes) *in vitro* grown mycelium. International Journal of Medicinal Mushrooms 13, 525–533.
Davoli, P., Mucci, A., Schenetti, L., Weber, R.W.S., 2005. Laetiporic acids, a family of non-carotenoid polyene pigments from fruit-bodies and liquid cultures of *Laetiporus sulphureus* (Polyporales, fungi). Phytochemistry 66, 817–823.
De Silva, D.D., Rapior, S., Fons, F., Bahkali, A.H., Hyde, K.D., 2012. Medicinal mushrooms in supportive cancer therapies: an approach to anti-cancer effects and putative mechanisms of action. Fungal Diversity 55, 1–35.
Demir, M.S., Yamac, M., 2008. Antimicrobial activities of basidiocarp, submerged mycelium and exopolysaccharide of some native basidiomycetes strains. Journal of Applied Biological Sciences 2, 89–93.
Du, D., Chen, Y., Chen, X., Wang, L., Feng, Y., Jiang, J., 2011. Analysis of chemical constituents of petroleum ether fraction from the fruiting body of *Fomes fomentarius* and its antitumor effect *in vitro*. Yaowu Fenxi Zazhi 31, 261–265.
Eghianruwa, Q., Odekanyin, O., Kuku, A., 2011. Physicochemical properties and acute toxicity studies of a lectin from the saline extract of the fruiting bodies of the shiitake mushroom, *Lentinula edodes* (Berk). International Journal of Biochemistry and Molecular Biology 2, 309–317.
Epstein, W.W., Sweat, F.W., Vanlear, G., Lovell, F.M., Gabe, E.J., 1979. Structure and stereochemistry of officinalic acid, a novel triterpene from *Fomes officinalis*. Journal of the American Chemical Society 101, 2748–2750.
Epstein, W.W., van Lear, G., 1966. Metabolites of *Fomes officinalis*. Journal of Organic Chemistry 31, 3434–3435.
Erb, B., Borschberg, H.J., Arigoni, D., 2000. The structure of laricinolic acid and its biomimetic transformation into officinalic acid. Journal of the Chemical Society, Perkin Transactions 1, 2307–2309.
Erkel, G., Anke, T., 2008. Products from basidiomycetes. Biotechnology Set, 2nd ed. Wiley, New York, pp. 490–526.
Fäldt, J., Jonsell, M., Nordlander, G., Borg-Karlson, A.-K., 1999. Volatiles of bracket fungi *Fomitopsis pinicola* and *Fomes fomentarius* and their functions as insect attractants. Journal of Chemical Ecology 25, 567–590.
Feng, W., Yang, J., 2010. Chemical constituents of *Fomes fomentarius* (L. Ex. Fr.). Zhongguo Yaoxue Zazhi 45, 1528–1530.
Feng, W., Yang, J., Xu, X., Liu, Q., 2010. Quantitative determination of lanostane triterpenes in *Fomes officinalis* and their fragmentation study by HPLC-ESI. Phytochemical Analysis 21, 531–538.
Ferreira, I.C., Vaz, J.A., Vasconcelos, M.H., Martins, A., 2010. Compounds from wild mushrooms with antitumor potential. Anticancer Agents in Medicinal Chemistry 10, 424–436.
Fischbein, C.B., Mueller, G.M., Leacock, P.R., Wahl, M.S., Aks, S.E., 2003. Digital imaging: a promising tool for mushroom identification. Academic Emergency Medicine 10, 808–811.
Gao, H.L., Lei, L.S., Yu, C.L., Zhu, Z.G., Chen, N.N., Wu, S.G., 2009. Immunomodulatory effects of *Fomes fomentarius* polysaccharides: an experimental study in mice. Nan Fang Yi Ke Da Xue Xue Bao 29, 458–461.
Garibay-Orijel, R., Caballero, J., Estrada-Torres, A., Cifuentes, J., 2007. Understanding cultural significance, the edible mushrooms case. Journal of Ethnobiology and Ethnomedicine 3, 1–18.
Gertsch, J., 2011. Botanical drugs, synergy, and network pharmacology: forth and back to intelligent mixtures. Planta Medica 77, 1086–1098.
Gonzaga, M.L., Bezerra, D.P., Alves, A.P., de Alencar, N.M., Mesquita Rde, O., Lima, M. W., Soares Sde, A., Pessoa, C., de Moraes, M.O., Costa-Lotufo, L.V., 2009. *in vivo* growth-inhibition of Sarcoma 180 by an alpha-(1,4)-glucan-beta-(1,6)-glucan-protein complex polysaccharide obtained from *Agaricus blazei* Murill. Journal of Natural Medicines 63, 32–40.
Göpfert, H., 1982. Pilze aus jungsteinzeitlichen Siedlungen. Schweizerische Zeitschrift für Pilzkunde, 50–70.
Q8
Grienke, U., Mihaly-Bison, J., Schuster, D., Afonyushkin, T., Binder, M., Guan, S.H., Cheng, C.R., Wolber, G., Stuppner, H., Guo, D.A., Bochkov, V.N., Rollinger, J.M., 2011. Pharmacophore-based discovery of FXR-agonists. Part II: identification of bioactive triterpenes from *Ganoderma lucidum*. Bioorganic & Medicinal Chemistry 19, 6779–6791.
Guler, P., Akata, I., Kutluer, F., 2009. Antifungal activities of *Fomitopsis pinicola* (Sw.: Fr) Karst and *Lactarius vellereus* (Pers.) Fr. African Journal of Biotechnology 8, 3811–3813.
Haarmann, T., Rolke, Y., Giesbert, S., Tudzynski, P., 2009. Ergot: from witchcraft to biotechnology. Molecular Plant Pathology 10, 563–577.
Härkönen, M., 2002. Mushroom Collecting in Tanzania and Hunan (Southern China); Inherited Wisdom and Folklore of Two Different Cultures. CABI Publishing, Wallingford.
Hartley, A.J., de Mattos-Shipley, K., Collins, C.M., Kilaru, S., Foster, G.D., Bailey, A.M., 2009. Investigating pleuromutilin-producing *Clitopilus* species and related basidiomycetes. FEMS Microbiology Letters 297, 24–30.
Hawksworth, D.L., 1996. Mycophobia and mycophilia. Nature 379, 503–504.
Herrmann, M., 1962. Die Verwendung des echten Zunderschwammes *Fomes fomentarius* einst und jetzt. Mycologisches Mitteilungsblatt 6, 56–62.
Hobbs, C., 1995. Medicinal Mushrooms: An Exploration of Tradition, Healing and Culture. Botanica Press, Santa Cruz.
Huang, T.-z., Du, D.-y., Chen, Y.-q., Yuan, B., Ju, X.-y., Feng, Y.-j., Wang, L., Jiang, J.-h., 2012. Chemical constituents and antitumor activity of fruiting body of *Fomes fomentarius*. Junwu Xuebao 31, 775–783.
Hwang, C.H., Jaki, B.U., Klein, L.L., Lankin, D.C., McAlpine, J.B., Napolitano, J.G., Fryling, N.A., Franzblau, S.G., Cho, S.H., Stamets, P.E., Wang, Y., Pauli, G.F., 2013. Chlorinated coumarins from the polypore mushroom *Fomitopsis officinalis* and their activity against *Mycobacterium tuberculosis*. Journal of Natural Products 76, 1916–1922.
Hwang, H.S., Yun, J.W., 2010. Hypoglycemic effect of polysaccharides produced by submerged mycelial culture of *Laetiporus sulphureus* on streptozotocin-induced diabetic rats. Biotechnology and Bioprocess Engineering 15, 173–181.
Hybelbauerova, S., Sejbal, J., Dracinsky, M., Hahnova, A., Koutek, B., 2008. Chemical constituents of *Stereum subtomentosum* and two other birch-associated basidiomycetes: an interspecies comparative study. Chemistry & Biodiversity 5, 743–750.
Ishii, P.L., Prado, C.K., Mauro, M.d.O., Carreira, C.M., Mantovani, M.S., Ribeiro, L.R., Dichi, J.B., Oliveira, R.J., 2011. Evaluation of *Agaricus blazei in vivo* for anti-genotoxic, anticarcinogenic, phagocytic and immunomodulatory activities. Regulatory Toxicology and Pharmacology 59, 412–422.
Jayasooriya, R.G.P.T., Kang, C.-H., Seo, M.-J., Choi, Y.H., Jeong, Y.-K., Kim, G.-Y., 2011. Exopolysaccharide of *Laetiporus sulphureus* var. *miniatus* downregulates LPS-induced production of NO, PGE(2), and TNF-alpha in BV2 microglia cells via suppression of the NF-kappa B pathway. Food and Chemical Toxicology 49, 2758–2764.
Jiang, M.H., Zhu, L., Jiang, J.G., 2010. Immunoregulatory actions of polysaccharides from Chinese herbal medicine. Expert Opinion on Therapeutic Targets 14, 1367–1402.
Jikai, L., 2002. Biologically active substances from mushrooms in Yunnan, China. Heterocycles 57, 157–167.
Jordan, M., 1995. Evidence of severe allergic reactions to *Laetiporus sulphureus*. Mycologist 9, 157–158.
Judova, J., Dubikova, K., Gaperova, S., Gaper, J., Pristas, P., 2012. The occurrence and rapid discrimination of *Fomes fomentarius* genotypes by ITS-RFLP analysis. Fungal Biology 116, 155–160.
Jung, M., Liermann, J.C., Opatz, T., Erkel, G., 2011. Ganodermycin, a novel inhibitor of CXCL10 expression from *Ganoderma applanatum*. Journal of Antibiotics 64, 683–686.
Kac, D., Barbieri, G., Falco, M.R., Seldes, A.M., Gros, E.G., 1984. The major sterols from 3 species of polyporaceae. Phytochemistry 23, 2686–2687.
Kamo, T., Asanoma, M., Shibata, H., Hirota, M., 2003. Anti-inflammatory lanostane-type triterpene acids from *Piptoporus betulinus*. Journal of Natural Products 66, 1104–1106.
Kandefer-Szerszeń, M., Kawecki, Z., 1974. Ether extracts from the fruiting body of *Piptoporus betulinus* as interference inducers. Acta Microbiologica Polonica. Series A: Microbiologia Generalis 6, 197–200.
Kandefer-Szerszen, M., Kawecki, Z., Guz, M., 1979. Fungal nucleic acids as interferon inducers. Biological Abstracts 95 (11)277–291 (Ref. 119022. 28).
Kang, C., Lee, C., Chung, K., Choi, E., Kim, B., 1982. An antitumor component of *Laetiporus sulphureus* and its immunostimulating activity. Archives of Pharmacal Research 5, 39–43.
Karaman, M., Jovin, E., Malbasa, R., Matavuly, M., Popovic, M., 2010. Medicinal and edible lignicolous fungi as natural sources of antioxidative and antibacterial agents. Phytotherapy Research 24, 1473–1481.
Karaman, M., Mimica-Dukic, N., Knezevic, P., Svircev, Z., Matavuly, M., 2009. Antibacterial properties of selected lignicolous mushrooms and fungi from Northern Serbia. International Journal of Medicinal Mushrooms 11, 269–279.
Kawagishi, H., Hamajima, K., Inoue, Y., 2002. Novel hydroquinone as a matrix metallo-proteinase inhibitor from the mushroom *Piptoporus betulinus*. Bioscience Biotechnology and Biochemistry 66, 2748–2750.
Keller, A.C., Maillard, M.P., Hostettmann, K., 1996. Antimicrobial steroids from the fungus *Fomitopsis pinicola*. Phytochemistry 41, 1041–1046.
Keller, C., Maillard, M., Keller, J., Hostettmann, K., 2002. Screening of European fungi for antibacterial, antifungal, larvicidal, molluscicidal, antioxidant and free-radical scavenging activities and subsequent isolation of bioactive compounds. Pharmaceutical Biology 40, 518–525.
Killermann, S., 1938. Ehemaliger Apothekerpilz. Zeitschrift für Pilzkunde 22, 11–13.
Kim, K.M., Lee, J.S., Jung, H.S., 2007. *Fomitopsis incarnatus* sp. nov. based on generic evaluation of *Fomitopsis* and *Rhodofomes*. Mycologia 99, 833–841.
Kim, K.M., Yoon, Y.G., Jung, H.S., 2005. Evaluation of the monophyly of *Fomitopsis* using parsimony and MCMC methods. Mycologia 97, 812–822.
Klaus, A., Kozarski, M., Niksic, M., Jakovljevic, D., Todorovic, N., Stefanoska, I., Van Griensven, L.J., 2013. The edible mushroom *Laetiporus sulphureus* as potential source of natural antioxidants. International Journal of Food Sciences and Nutrition 64, 599–610.
Konska, G., Guillot, J., Dusser, M., Damez, M., Botton, B., 1994. Isolation and characterization of an N-acetyllactosamine-binding lectin from a mushroom *Laetiporus sulfureus*. Journal of Biochemistry 116, 519–523.
Kreisel, H., 1997. Jesus, Petrus und die Entstehung der Pilze. Tintling 1, 29–30.
Kreisel, H., 1998. Mushrooms in painting. European Council for Conservation of Fungi (ECCF).
Krzyczkowski, W., Malinowska, E., Suchocki, P., Kleps, J., Olejnik, M., Herold, F., 2009. Isolation and quantitative determination of ergosterol peroxide in various edible mushroom species. Food Chemistry 113, 351–355.
Lear, M.J., Simon, O., Foley, T.L., Burkart, M.D., Baiga, T.J., Noel, J.P., DiPasquale, A.G., Rheingold, A.L., La Clair, J.J., 2009. Laetirobin from the parasitic growth of *Laetiporus sulphureus* on *Robinia pseudoacacia*. Journal of Natural Products 72, 1980–1987.
Lee, J.-S., 2005. Effects of *Fomes fomentarius* supplementation on antioxidant enzyme activities, blood glucose, and lipid profile in streptozotocin-induced diabetic rats. Nutrition Research 25, 187–195.

UNCORRECTED PROOF

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Lee, S.-I., Kim, J.-S., Oh, S.-H., Park, K.-Y., Lee, H.-G., Kim, S.-D., 2008. Antihyperglycemic effect of *Fomitopsis pinicola* extracts in streptozotocin-induced diabetic rats. Journal of Medicinal Food 11, 518–524.

Leeser, O., 1987. Lehrbuch der Homöopathie. 1973 karl F. Haug Verlag, Heidelberg, Heidelberg.

Lemieszek, M.K., Langner, E., Kaczor, J., Kandefer-Szerszen, M., Sanecka, B., Mazurkiewicz, W., Rzeski, W., 2009. Anticancer effect of fraction isolated from medicinal birch polypore mushroom, *Piptoporus betulinus* (Bull.: Fr.) P. Karst. (Aphyllophoromycetideae): *in vitro* studies. International Journal of Medicinal Mushrooms 11, 351–364.

León, F., Quintana, J., Rivera, A., Estevez, F., Bermejo, J., 2004. Lanostanoid triterpenes from *Laetiporus sulphureus* and apoptosis induction on HL-60 human myeloid leukemia cells. Journal of Natural Products 67, 2008–2011.

Leonowicz, A., Matusjewska, A., Luterek, J., Ziegenhagen, D., Wojtas-Wasilewska, M., Cho, N.-S., Hofrichter, M., Roglaski, J., 1999. Biodegradation of lignin by white rot fungi. Fungal Genetics and Biology 27, 175–185.

Lindequist, U., Rausch, R., Fussel, A., Hanssen, H.P., 2010. Higher fungi in traditional and modern medicine. Medizinische Monatsschrift für Pharmazeuten 33, 40–48.

Lindner, D.L., Banik, M.T., 2008. Molecular phylogeny of *Laetiporus* and other brown rot polypore genera in North American Mycologia 100, 417–430.

List, P.H., Menssen, H.G., 1959. Basic fungal substances. III. Volatile amines and amino acids of *Polyporus sulphureus* Bull. Archiv der Pharmazie und Berichte der Deutschen Pharmazeutischen Gesellschaft 292/64, 21–28.

Liu, X.-T., Winkler, A.L., Schwan, W.R., Volk, T.J., Rott, M., Monte, A., 2010. Antibacterial compounds from mushrooms II: Lanostane triterpenoids and an ergostane steroid with activity against *Bacillus cereus* isolated from *Fomitopsis pinicola*. Planta Medica 76, 464–466.

Lohwag, K., 1965. Birkenschwammschnitzereien aus der Steiermark. Mitteilungsblatt des Naturwissenschaftlichen Vereins für Steiermark 95, 136–139.

Lucas, E.H., 1960. Folklore and Plant Drugs. Papers of the Michigan Academy of Sience, Arts, and LettersXLV, 127–136.

Macakova, K., Opletal, L., Polásek, M., Samkova, V., 2010. Free-radical scavenging activity of some European polyporales. Natural Product Communications 5, 923–926.

Marsh, R.W., 1973. Micological millinery. Bulletin of the British Mycological Society 7, 35.

Marzell, H., 1921. Die Pilze im Volksaberglauben. Pilz und Kräuterfreund 5, 59–60.

Matt, G., 1947. Pilze heilen Rheumatismus. Schweizerische Zeitschrift für Pilzkunde 25, 167–168.

Mattila, P., Suonpää, K., Piironen, V., 2000. Functional properties of edible mushrooms. Nutrition 16, 694–696.

McCormick, M.A., Grand, L.F., Post, J.B., Cubeta, M.A., 2013. Phylogenetic and phenotypic characterization of *Fomes fasciatus* and *Fomes fomentarius* in the United States. Mycologia 105, 1524–1534.

Q9 Mellin, C.J., 1791. Die Hausmitttel. Graz.

Meng, L.-Z., Lin, B.-Q., Wang, B., Feng, K., Hu, D.-J., Wang, L.-Y., Cheong, K.-L., Zhao, J., Li, S.-P., 2013. Mycelia extracts of fungal strains isolated from *Cordyceps sinensis* differently enhance the function of RAW 264.7 macrophages. Journal of Ethnopharmacology 148, 818–825.

Mo, H., Winter, H.C., Goldstein, I.J., 2000. Purification and characterization of a Neu5Acα2-6Galβ1-4Glc/GlcNAc-specific lectin from the fruiting body of the polypore mushroom *Polyporus squamosus*. Journal of Biological Chemistry 275, 10623–10629.

Molitoris, H.P., 2002. Pilze in Medizin, Folklore und Religion. Feddes Repert 113, 165–182.

Molitoris, H.P., 2005. Fungi: companions of Man in Good and Evil. International Journal Medicinal Mushrooms 7, 49–73.

Montoya, A., Kong, A., Estrada-Torres, A., Cifuentes, J., Caballero, J., 2004. Useful wild fungi of La Malinche National Park, Mexico. Fungal Diversity 17, 115–143.

Moradali, M.F., Mostafavi, H., Ghods, S., Hedjaroude, G.A., 2007. Immunomodulating and anticancer agents in the realm of macromycetes fungi (macrofungi). International Immunopharmacology 7, 701–724.

Moro, C., Palacios, I., Lozano, M., D'Arrigo, M., Guillamon, E., Villares, A., Martinez, J. A., Garcia-Lafuente, A., 2012. Anti-inflammatory activity of methanolic extracts from edible mushrooms in LPS activated RAW 264.7 macrophages. Food Chemistry 130, 350–355.

Muhsin, T.M., Al Duboon, A.A., Khalaf, K.T., 2011. Bioactive compounds from a polypore fungus *Ganoderma applanatum* (Pers ex Wallr.). Jordan Journal of Biological Sciences 4, 205–212.

Nam, K.S., Jo, Y.S., Kim, Y.H., Hyun, J.W., Kim, H.W., 2001. Cytotoxic activities of acetoxyscirpenediol and ergosterol peroxide from *Paecilomyces tenuipes*. Life Sciences 69, 229–237.

Nicholson, M.S., 2009. Some Spiritual Uses of Mushrooms. Fungi 2, 23–27.

Olennikov, D.N., Agafonova, S.V., Borovskii, G.B., Penzina, T.A., Rokhin, A.V., 2009a. Alkali-soluble polysaccharides of *Laetiporus sulphureus* (Bull.: Fr.) Murr fruit bodies. Applied Biochemistry and Microbiology 45, 626–630.

Olennikov, D.N., Agafonova, S.V., Borovskii, G.B., Penzina, T.A., Rokhin, A.V., 2009b. Water-soluble endopolysaccharides from the fruiting bodies of *Laetiporus sulphureus* (Bull.: Fr.) Murr. Applied Biochemistry and Microbiology 45, 536–543.

Olennikov, D.N., Agafonova, S.V., Nazarova, A.V., Borovskii, G.B., Penzina, T.A., 2008. Organic acids and carbohydrates from *Laetiporus sulphureus* fruiting bodies. Chemistry of Natural Compounds 44, 762–763.

Olennikov, D.N., Tankhaeva, L.M., Agafonova, S.V., 2011. Antioxidant components of *Laetiporus sulphureus* (Bull.: Fr.) Murr. fruit bodies. Applied Biochemistry and Microbiology 47, 419–425.

Orhan, I., Üstün, O., 2011. Determination of total phenol content, antioxidant activity and acetylcholinesterase inhibition in selected mushrooms from Turkey. Journal of Food Composition and Analysis 24, 386–390.

Ota, Y., Hattori, T., Banik, M.T., Hagedorn, G., Sotome, K., Tokuda, S., Abe, Y., 2009. The genus *Laetiporus* (Basidiomycota, Polyporales) in East Asia. Mycological Research 113, 1283–1300.

Ozen, T., Darcan, C., Aktop, O., Turkekul, I., 2011. Screening of antioxidant, antimicrobial activities and chemical contents of edible mushrooms wildly grown in the Black Sea region of Turkey. Combinatorial Chemistry & High Throughput Screening 14, 72–84.

Park, Y.M., Kim, I.T., Park, H.J., Choi, J.W., Park, K.Y., Lee, J.D., Nam, B.H., Kim, D.G., Lee, J.Y., Lee, K.T., 2004. Anti-inflammatory and anti-nociceptive effects of the methanol extract of *Fomes fomentarius*. Biological & Pharmaceutical Bulletin 27, 1588–1593.

Paterson, R.R.M., 2006. *Ganoderma* – a therapeutic fungal biofactory. Phytochemistry 67, 1985–2001.

Pegler, D.N., 2000. Useful fungi of the world: some use of bracket fungi. Mycologist 14, 6–7.

Peintner, U., Pöder, R., 2000. Ethnomycological remarks on the Iceman's fungi. In: Bortenschlager, S., Oeggl, K. (Eds.), The Iceman and his Natural Environment. Springer, Vienna, pp. 143–150.

Peintner, U., Pöder, R., Pümpel, T., 1998. The Iceman's fungi. Mycological Research 102, 1153–1162.

Peintner, U., Schwarz, S., Mesic, A., Moreau, P.-A., Moreno, G., Saviuc, P., 2013. Mycophilic or Mycophobic? Legislation and guidelines on wild mushroom commerce reveal different consumption behaviour in European countries. PLoS ONE 8, e63926.

Peng, Y., Yang, X., Zhang, Y., 2005. Microbial fibrinolytic enzymes: an overview of source, production, properties, and thrombolytic activity *in vivo*. Applied Microbiology and Biotechnology 69, 126–132.

Petrova, A., Popov, S., Gjosheva, M., Bankova, V., 2007. A new triterpenic alcohol from *Fomitopsis pinicola*. Natural Product Research 21, 401–405.

Petrova, R.D., Reznick, A.Z., Wasser, S.P., Denchev, C.M., Nevo, E., Mahajna, J., 2008. Fungal metabolites modulating NF-kappa B activity: an approach to cancer therapy and chemoprevention. Oncology Reports 19, 299–308.

Pöder, R., Peintner, U., 1999. Laxatives and the Ice Man. Lancet 353, 926. Q10

Rapior, S., Konska, G., Guillot, J., Andary, C., Bessiere, J.M., 2000. Volatile composition of *Laetiporus sulphureus*. Cryptogamie Mycology 21, 67–72.

Ren, G., Liu, X.Y., Zhu, H.K., Yang, S.Z., Fu, C.X., 2006. Evaluation of cytotoxic activities of some medicinal polypore fungi from China. Fitoterapia 77, 408–410.

Rios, J.-L., Andujar, I., Recio, M.-C., Giner, R.-M., 2012. Lanostanoids from fungi: a group of potential anticancer compounds. Journal of Natural Products 75, 2016–2044.

Rösecke, J., König, W.A., 1999. Steroids from the fungus *Fomitopsis pinicola*. Phytochemistry 52, 1621–1627.

Rösecke, J., König, W.A., 2000. Constituents of various wood-rotting basidiomycetes. Phytochemistry 54, 603–610.

Roussel, B., Rapior, S., Masson, C.L., Boutie, P., 2002. *Fomes fomentarius* (L.: Fr.) Fr.: a mushroom with varied uses. Cryptogamie Mycology 23, 349–366.

Rutalek, R., 2002. Ethnomykologie – Eine Übersicht. Österreichische Zeitschrift für Pilzkunde 11, 79–94.

Saar, M., 1991. Fungi in Khanty folk medicine. Journal of Ethnopharmacology 31, 175–179.

Sakagami, H.A.T., 1991. Induction of immunopotentiation activity by a protein bound polysaccharide (PSK). Anticancer Research 11, 993–1000. Q11

Santana, M.N., Oliveira, G.C.M., Yoshida, V.M.H., Sabha, M., Oshima-Franco, Y., 2011. Naturally occurring ingredients as potential antiaging cosmetics. Latin American Journal of Pharmacy 30, 1531–1535.

Schlegel, B., Luhmann, U., Hartl, A., Grafe, U., 2000. Piptamine, a new antibiotic produced by *Piptoporus betulinus* Lu 9-1. Journal of Antibiotics 53, 973–974.

Schoch, C.L., Seifert, K.A., Huhndorf, S., Robert, V., Spouge, J.L., Levesque, C.A., Chen, W., Bolchacova, E., Voigt, K., Crous, P.W., Miller, A.N., Fungal Barcoding, C., et al., 2012. Nuclear ribosomal internal transcribed spacer (ITS) region as a universal DNA barcode marker for fungi. Proceedings of the National Academy of Science of the United States of America 109, 6241–6246.

Seniuk, O.F., Gorovoj, L.F., Beketova, G.V., Savichuk, H.O., Rytik, P.G., Kucherov, I.I., Prilutskay, A.B., Prilutsky, A.I., 2011. Anti-infective properties of the melanin-glucan complex obtained from medicinal tinder bracket mushroom, *Fomes fomentarius* (L.: Fr.) Fr. (Aphyllophoromycetideae). International Journal of Medicinal Mushrooms 13, 7–18.

Senn-Irlet, B., 2012. Merkblatt Pilze: Lärchenschwamm. *Laricifomes officinalis* (Vill.) Kotl. und Pouz. Eidg. Forschungsanstalt WSL Birmensdorf.

Seo, M.J., Kang, B.W., Park, J.U., Kim, M.J., Lee, H.H., Choi, Y.H., Jeong, Y.K., 2011. Biochemical characterization of the exopolysaccharide purified from *Laetiporus sulphureus* mycelia. Journal of Microbiology and Biotechnology 21, 1287–1293.

Shashkina, M.Y., Shashkin, P.N., Sergeev, A.V., 2006. Chemical and medicobiological properties of chaga. Pharmaceutical Chemistry Journal 40, 560–568.

Sheth, K., Catafomo, P., Sciuchetti, L.A., 1967. Isolation and identification of eburicoic acid from *Fomes pinicola*. Journal of Pharmaceutical Sciences 56, 1656–1658.

Shiono, Y., Tamesada, Y., Muravayev, Y.D., Murayama, T., Ikeda, M., 2005. N-Phenethylhexadecanamide from the edible mushroom *Laetiporus sulphureus*. Natural Product Research 19, 363–366.

Shu, C.H.L., Lin, K.J., Wen, B.J., 2004. Effects of culture pH on the production of bioactive polysaccharides by *Agaricus blazei* in batch cultures. Journal of Chemical Technology and Biotechnology 79, 998–1002.

UNCORRECTED PROOF

ARTICLE IN PRESS

Sidorenko, M.L., Buzoleva, L.S., 2012. Search for new types of raw materials for antibacterial drugs. Antibiotiki i khimioterapiia=Antibiotics and chemoterapy [sic]/Ministerstvo meditsinskoi i mikrobiologicheskoi promyshlennosti SSSR 57, 7–10.

Smith, H.I., 1929. Materia medica of the Bella Coola and neighbouring tribes of British Columbia. In: Printer, K. (Ed.), Annual Report. National Museum of Canada Bulleti, Ottawa, Canada.

Sporenheimer, B., 1936. Eine ungewöhnliche große Nebennutzung aus dem Baumschwamm. Zeitschrift für Pilzkunde 20, 77–79.

Sulkowska-Ziaja, K., Muszynska, B., Motyl, P., Pasko, P., Ekiert, H., 2012. Phenolic compounds and antioxidant activity in some species of polyporoid mushrooms from Poland. International Journal of Medicinal Mushrooms 14, 385–393.

Sun, X., Zhao, X.-h., Bao, H.-y., 2012. Antitumor active constituent in fruiting body of *Fomitopsis pinicola*. Shizhen Guoyi Guoyao 23, 1634–1637.

Szedlay, G., Jakucs, E., Boldizsar, I., Boka, K., 1999. Basidiocarp and mycelium morphology of *Ganoderma lucidum* Karst. strains isolated in Hungary. Acta Microbiologica et Immunologica Hungarica 46, 41–52.

Takemoto, T., Nakajima, T., Sakuma, P., 1964. Isolation of a flycidal constituent "Ibotenic acid" from *Amanita muscaria* and *A. pantherina*. Yakugaku Zasshi 84, 1233–1234.

Teplyakova, T.V., Psurtseva, N.V., Kosogova, T.A., Mazurkova, N.A., Khanin, V.A., Vlasenko, V.A., 2012. Antiviral activity of polyporoid mushrooms (higher basidiomycetes) from Altai mountains (Russia). International Journal of Medicinal Mushrooms 14, 37–45.

Q12 Thoen, D., 1982. Usage et legendes lies aux polypores. BSMF 98 (3), 289–318.

Thyagarajan-Sahu, A., Lane, B., Sliva, D., 2011. ReishiMax, mushroom based dietary supplement, inhibits adipocyte differentiation, stimulates glucose uptake and activates AMPK. BMC Complementary Alternative Medicine 11, 74.

Turkoglu, A., Duru, M.E., Mercan, N., Kivrak, I., Gezer, K., 2007. Antioxidant and antimicrobial activities of *Laetiporus sulphureus* (Bull.) Murrill. Food Chemistry 101, 267–273.

Utzig, J.S.Z., 1957. Wpływ trójterpenów zawartych w żagwi brzozowej – *Polyporus betulinus* na guzy. Stickera MedWet 8, 481–484.

Vaidya, J.G., Rabba, A.S., 1993. Fungi in folk medicine. Mycologist 7, 131–133.

Vannucci, L., Krizan, J., Sima, P., Stakheev, D., Caja, F., Rajsiglova, L., Horak, V., Saieh, M., 2013. Immunostimulatory properties and antitumor activities of glucans. International Journal of Oncology 43, 357–364.

Vasaitis, R., Menkis, A., Lim, Y.W., Seok, S., Tomsovsky, M., Jankovsky, L., Lygis, V., Slippers, B., Stenlid, J., 2009. Genetic variation and relationships in *Laetiporus sulphureus* s. lat., as determined by ITS rDNA sequences and *in vitro* growth rate. Mycological Research 113, 326–336.

Vieira, G.R.T., Liebl, M., Tavares, L.B.B., Paulert, R., Smânia Júnior, A., 2008. Submerged culture conditions for the production of mycelial biomass and antimicrobial metabolites by *Polyporus tricholoma* Mont. Brazilian Journal of Microbiology 39, 561–568.

Wangun, H.V.K., Berg, A., Hertel, W., Nkengfack, A.E., Hertweck, C., 2004. Anti-inflammatory and anti-hyaluronate lyase activities of lanostanoids from *Piptoporus betulinus*. Journal of Antibiotics 57, 755–758.

Wasser, S.P., Weis, A.L., 1999. Therapeutic effects of substances occurring in higher basidiomycetes mushrooms: a modern perspective. Critical Reviews in Immunology 19, 65–96.

Wasson, R.G., 1969. Soma: Divine Mushroom of Immortality. Harcourt, Brace & World, New York.

Wasson, V.P., Wasson, R.G., 1957. Mushrooms, Russia, and history. Pantheon Books, New York.

Watling, R., 1997. Poisoning by fungi: interesting cases. Mycologist 11, 101.

Weber, F.C., 1958. Pilze als Heilmittel. Schweizer Zeitschrift für Pilzkunde 35, 90–93.

Weber, R.W.S., Mucci, A., Davoli, P., 2004. Laetiporic acid, a new polyene pigment from the wood-rotting basidiomycete *Laetiporus sulphureus* (Polyporales, Fungi). Tetrahedron Letters 45, 1075–1078.

Weng, C.J., Yen, G.C., 2010. The *in vitro* and *in vivo* experimental evidences disclose the chemopreventive effects of *Ganoderma lucidum* on cancer invasion and metastasis. Clinical and Experimental Metastasis 27, 361–369.

Wiater, A., Paduch, R., Pleszczynska, M., Prochniak, K., Choma, A., Kandefer-Szerszen, M., Szczodrak, J., 2011. alpha-(1,3)-D-glucans from fruiting bodies of selected macromycetes fungi and the biological activity of their carboxymethylated products. Biotechnology Letters 33, 787–795.

Wiater, A., Pleszczynska, M., Szczodrak, J., Janusz, G., 2012. Comparative studies on the induction of *Trichoderma harzianum* mutanase by alpha-(1,3)-glucan-rich fruiting bodies and mycelia of *Laetiporus sulphureus*. International Journal of Molecular Sciences 13, 9584–9598.

Wu, S.M., Zorn, H., Krings, U., Berger, R.G., 2005. Characteristic volatiles from young and aged fruiting of wild *Polyporus sulfureus* (Bull.: Fr.) fr. Journal of Agricultural and Food Chemistry 53, 4524–4528.

Wu, X., Yang, J.-S., Yan, M., 2009. Four New Triterpenes from Fungus of *Fomes officinalis*. Chemical and Pharmaceutical Bulletin 57, 195–197.

Wu, X., Yang, J., Dong, Y., 2005. Chemical constituents of *Fomes officinalis* (I). Zhongcaoyao 36, 811–814.

Wu, X., Yang, J.S., Zhou, L., Dong, Y.S., 2004. New lanostane-type triterpenes from *Fomes officinalis*. Chemical and Pharmaceutical Bulletin 52, 1375–1377.

Xu, X., Wu, Y., Chen, H., 2011a. Comparative antioxidative characteristics of polysaccharide-enriched extracts from natural sclerotia and cultured mycelia in submerged fermentation of *Inonotus obliquus*. Food Chemistry 127, 74–79.

Xu, X., Yan, H., Chen, J., Zhang, X., 2011b. Bioactive proteins from mushrooms. Biotechnology Advances 29, 667–674.

Ying, J., Mao, X., Ma, Q., Zong, Y., Wen, H., 1987. Icones of Medical Fungi From China. Science Press, Beijing.

Yoshikawa, K., Bando, S., Arihara, S., Matsumura, E., Katayama, S., 2001. A benzofuran glycoside and an acetylenic acid from the fungus *Laetiporus sulphureus* var. *miniatus*. Chemical and Pharmaceutical Bulletin 49, 327–329.

Yoshikawa, K., Inoue, M., Matsumoto, Y., Sakakibara, C., Miyataka, H., Matsumoto, H., Arihara, S., 2005. Lanostane triterpenoids and triterpene glycosides from the fruit body of *Fomitopsis pinicola* and their inhibitory activity against COX-1 and COX-2. Journal of Natural Products 68, 69–73.

Zang, Y., Xiong, J., Zhai, W.Z., Cao, L., Zhang, S.P., Tang, Y., Wang, J., Su, J.J., Yang, G.X., Zhao, Y., Fan, H., Xia, G., Wang, C.G., Hu, J.F., 2013. Fomentarols A-D, sterols from the polypore macrofungus *Fomes fomentarius*. Phytochemistry 92, 137–145.

Zhang, H., Feng, B., Liu, K., Guo, S., 2013. Isolation and purification of two triterpenoids from the Chinese medicinal plant *Fomes officinalis* ames. Asian Journal of Chemistry 25, 6130–6132.

Zhang, Y., Zhao, Y., Cui, H., Cao, C., Guo, J., Liu, S., 2011. Comparison of hypoglycemic activity of fermented mushroom of *Inonotus obliquus* rich in vanadium and wild-growing *I. obliquus*. Biological Trace Element Research 144, 1351–1357.

Zhong, J.-J., Tang, Y.-J., 2004. Submerged cultivation of medicinal mushrooms for production of valuable bioactive metabolites. In: Zhong, J.-J. (Ed.), Biomanufacturing. Springer, Berlin Heidelberg, pp. 25–59.

Zhong, X.-h., Ren, K., Lu, S.-j., Yang, S.-y., Sun, D.-z., 2009. Progress of research on *Inonotus obliquus*. Chinese Journal of Integrative Medicine 15, 156–160.

Zjawiony, J.K., 2004. Biologically active compounds from *Aphyllophorales* (polypore) fungi. Journal of Natural Products 67, 300–310.

UNCORRECTED PROOF

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